

Lecture 10

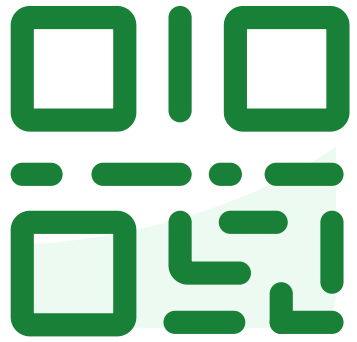
Logistic Regression (1)

Introduction to Logistic Regression

EECS 189/289, Fall 2025 @ UC Berkeley

Joseph E. Gonzalez and Narges Norouzi

Do not edit
How to change the
design



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Roadmap

- Classification Task
- Defining a New Model for Classification
- Linear Discriminant Functions
- Discriminative Probabilistic Models
- Sigmoid for Classification
- Logistic Regression Objective
- Regularization with Logistic Regression
- Logistic Regression Optimization
- Making Decisions



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Classification Task

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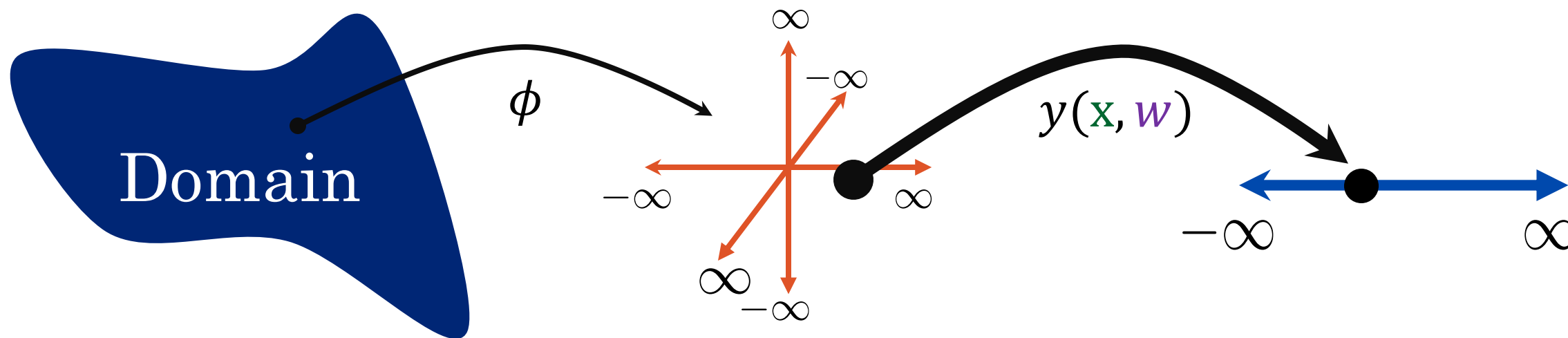


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So Far



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Squared Error

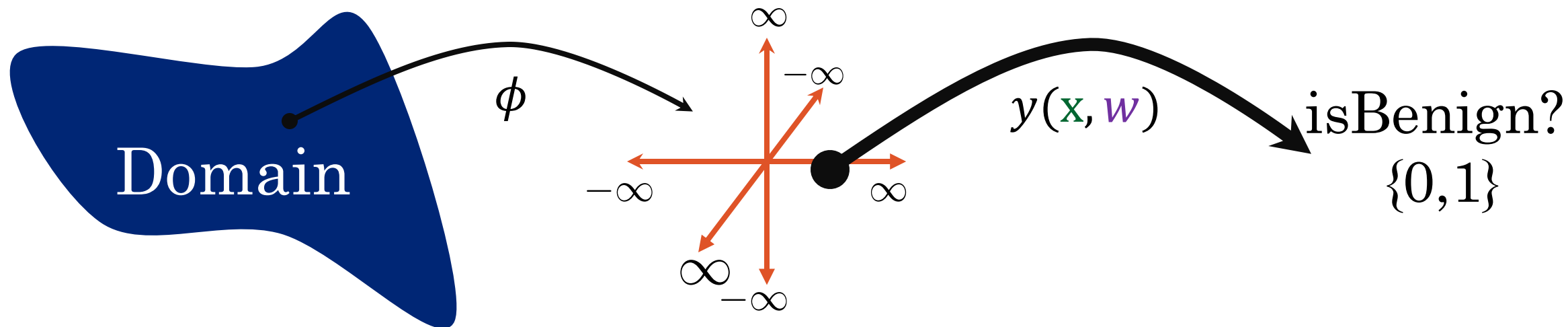
$$\operatorname{argmin}_{\mathbf{w}} \frac{1}{2} \sum_{n=1}^N \left(t_n - \mathbf{w}^T \phi(\mathbf{x}_n) \right)^2 + \frac{\lambda}{2} \mathbf{w}^T \mathbf{w}$$

Regularization

Classification Task Shift



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Kinds of Classification



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We want to predict some **categorical variable**, or **response**, y .

- **Binary** classification

- Two classes
- **Responses** (y) are either 0 or 1

win or lose

disease or
no disease

- **Multiclass** classification

- Many classes
- Image labeling and sentiment analysis (positive, negative, or neutral).



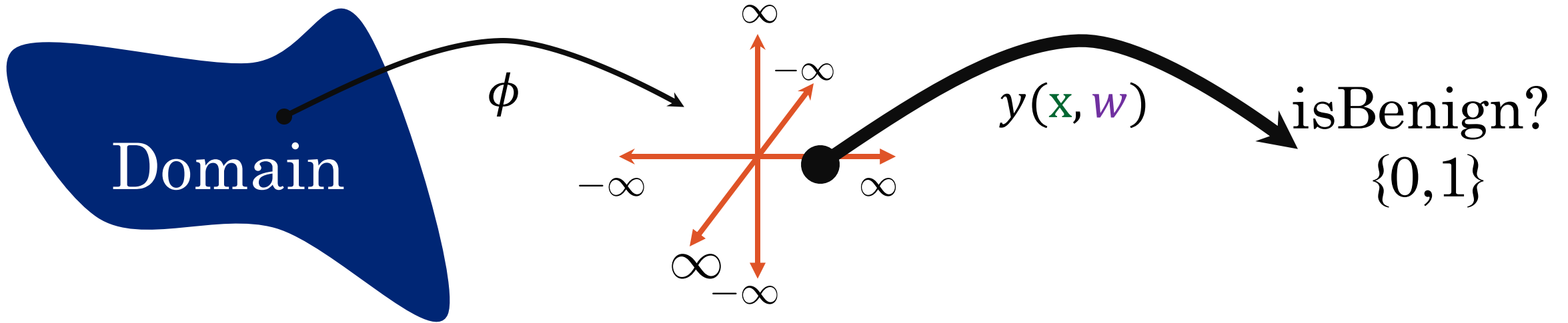
- **Structured prediction** tasks

- Predicting a structured object instead of a discrete class
- Examples: Translation and ChatGPT.

Classification



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Can we just use least squares?

$$\operatorname{argmin}_{\mathbf{w}} \frac{1}{2} \sum_{n=1}^N \left(t_n - \mathbf{w}^T \phi(\mathbf{x}_n) \right)^2 + \frac{\lambda}{2} \mathbf{w}^T \mathbf{w}$$

Squared Error

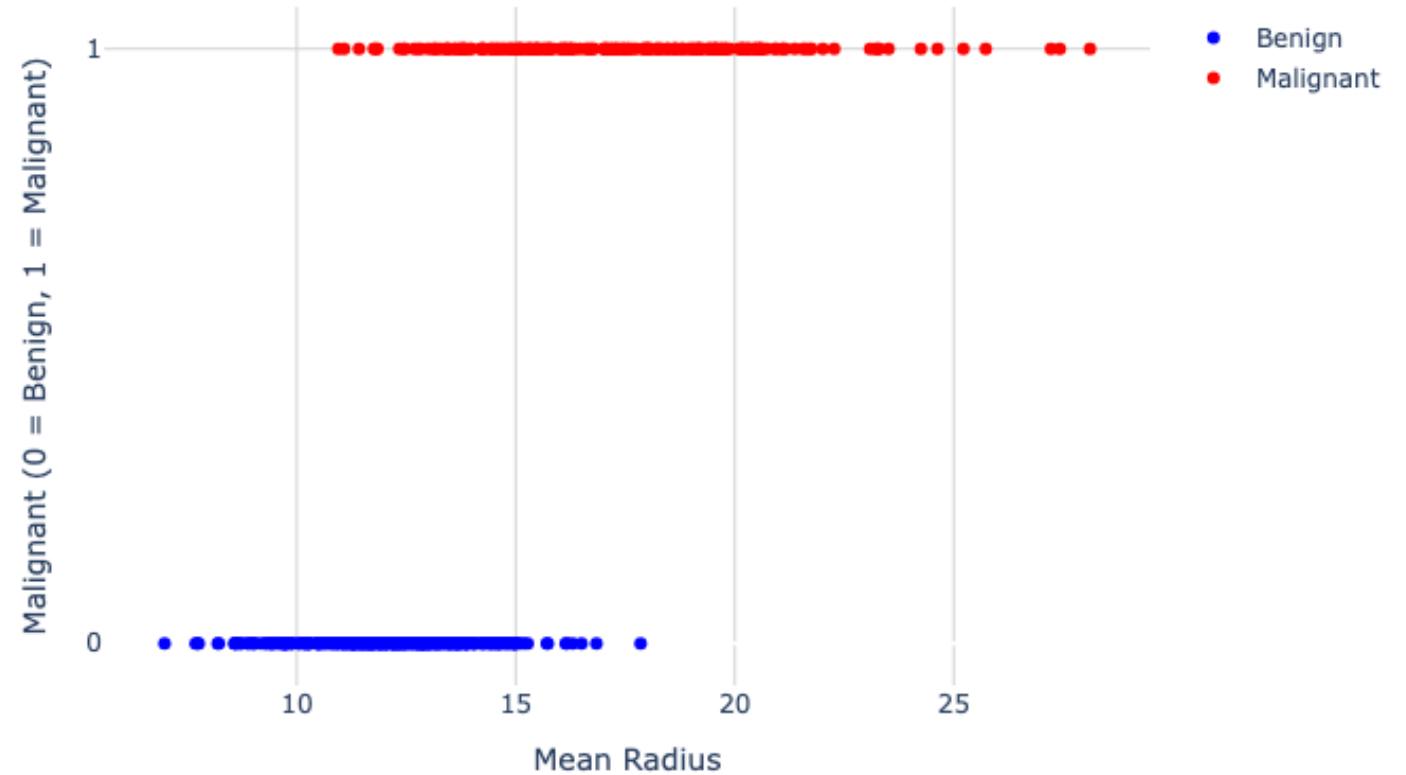
Regularization



Demo

We see the distribution of the two classes in the figure. Y axis shows the class of the image (benign or malignant) and the X axis shows Mean Radius.

The Distribution of the Data Between the Two Classes



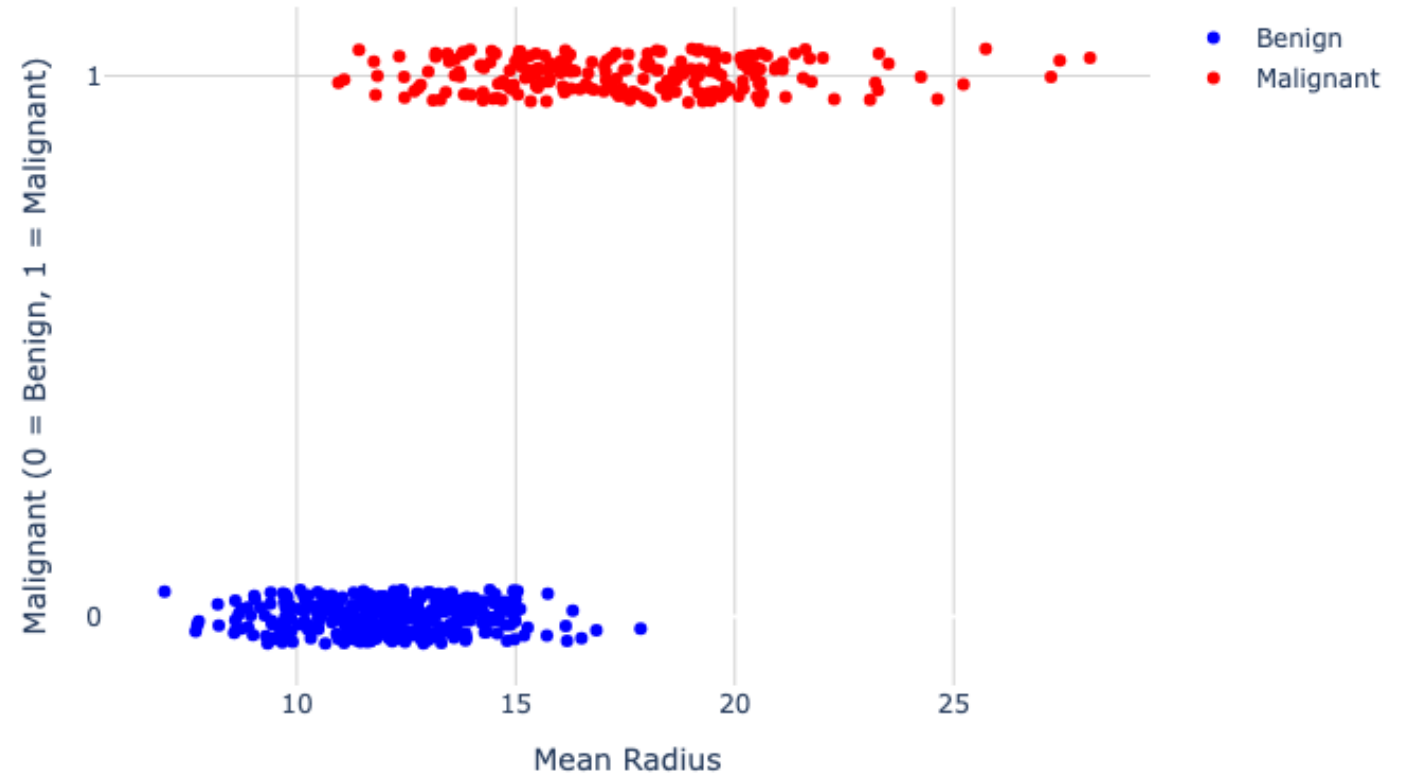


Demo

We usually add jittering to be able to see all data points clearly.

```
def jitter(data, amt=0.1):  
    return data +  
        amt*np.random.rand(len(data))  
        - amt/2.0
```

The Distribution of the Data Between the Two Classes (Jittered)

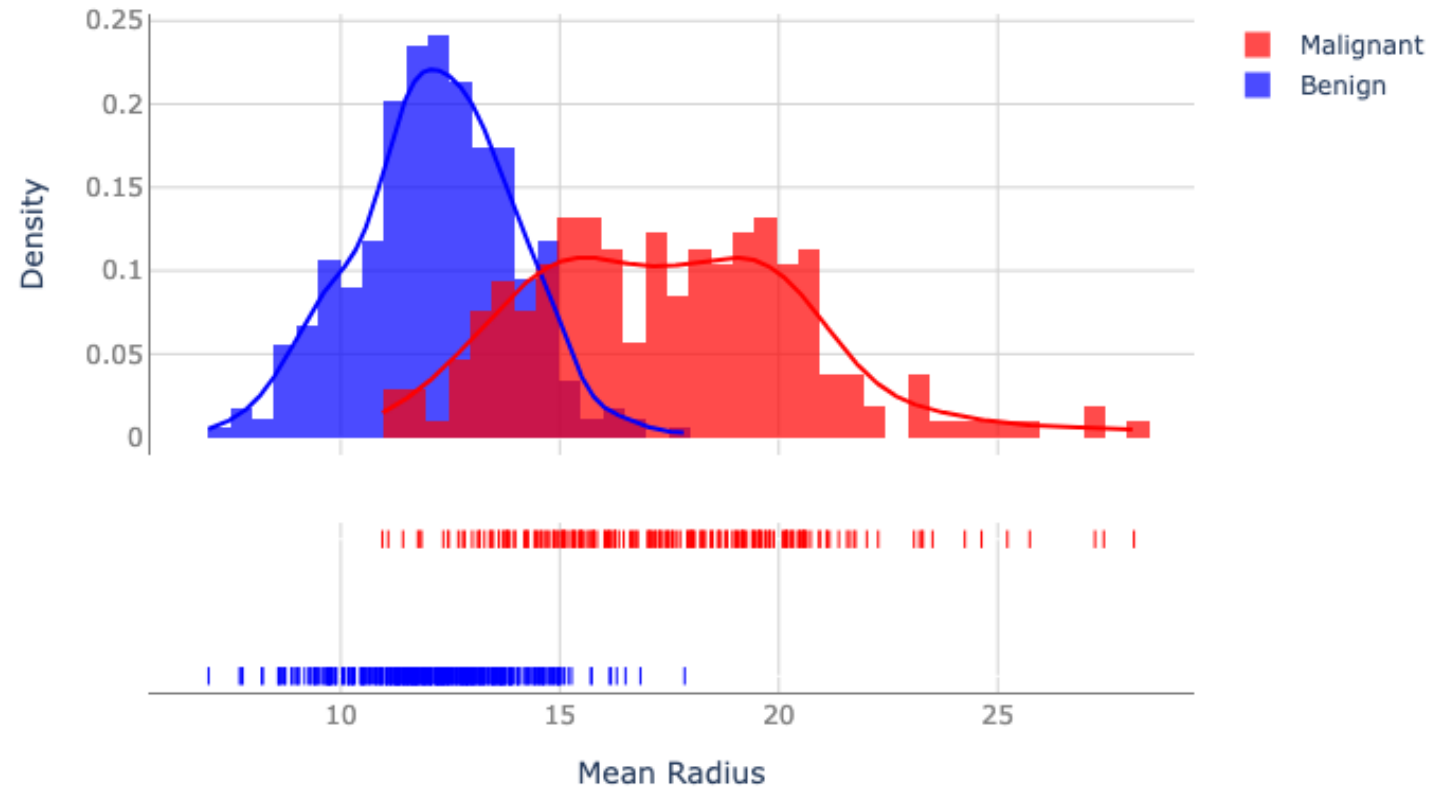




Demo

Here is the probability density of the data for both classes.

The Probability Density of the Data



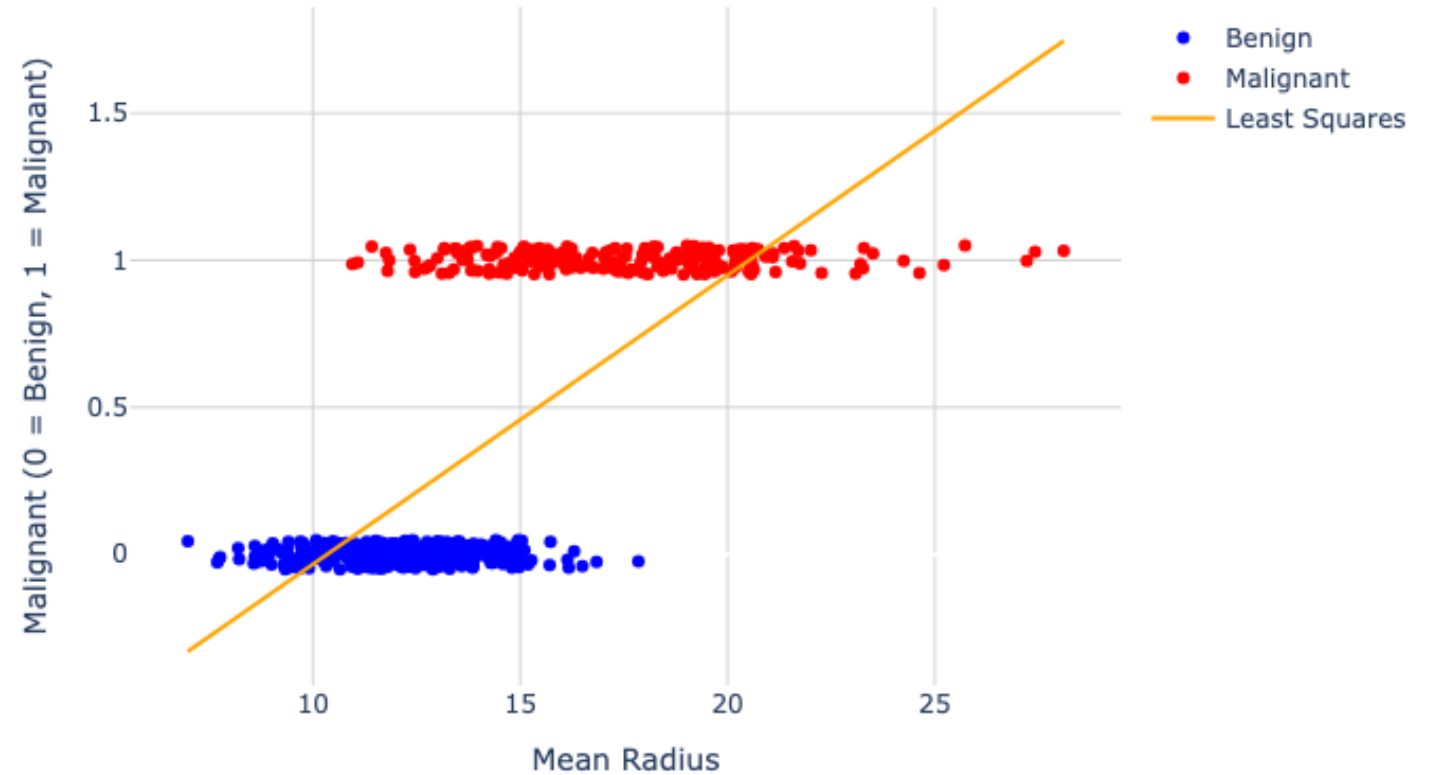


Demo

What if we test linear regression model for this task?

Here, the orange line is a regression line fit to the Mean Radius feature, mapping it to the scalar target values (0s and 1s).

Linear Regression Fitted to the Dataset



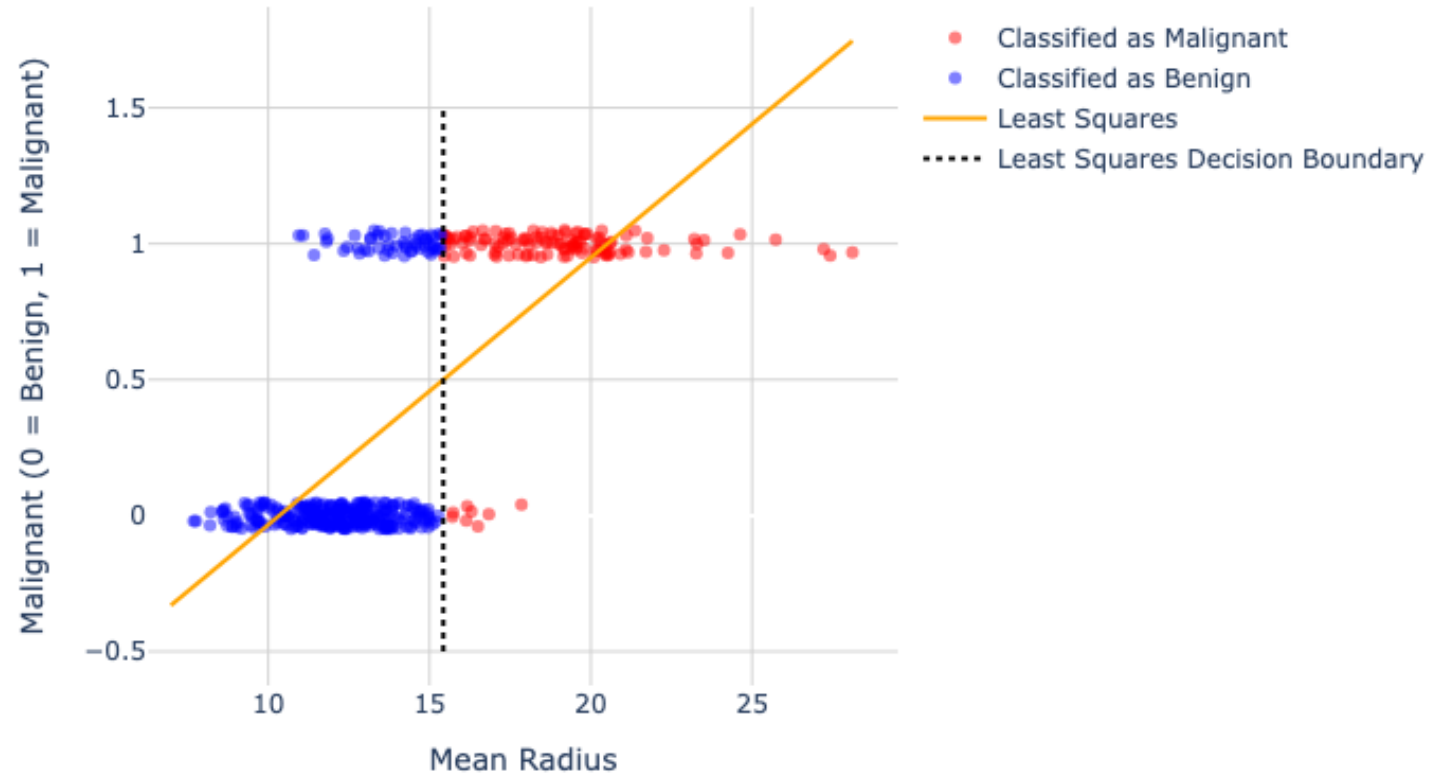


Demo

We needed a decision function
(e.g., $f(x) > 0.5$) ...

If we use the threshold of 0.5
between the two target values 0
and 1 and highlight all data points
that are labeled as benign with
blue and all malignant predictions
with red, you will see that a
decision boundary is being formed
between the two classes with the
black dashed line.

Use of 0.5 as the Decision Boundary Between the Two Classes



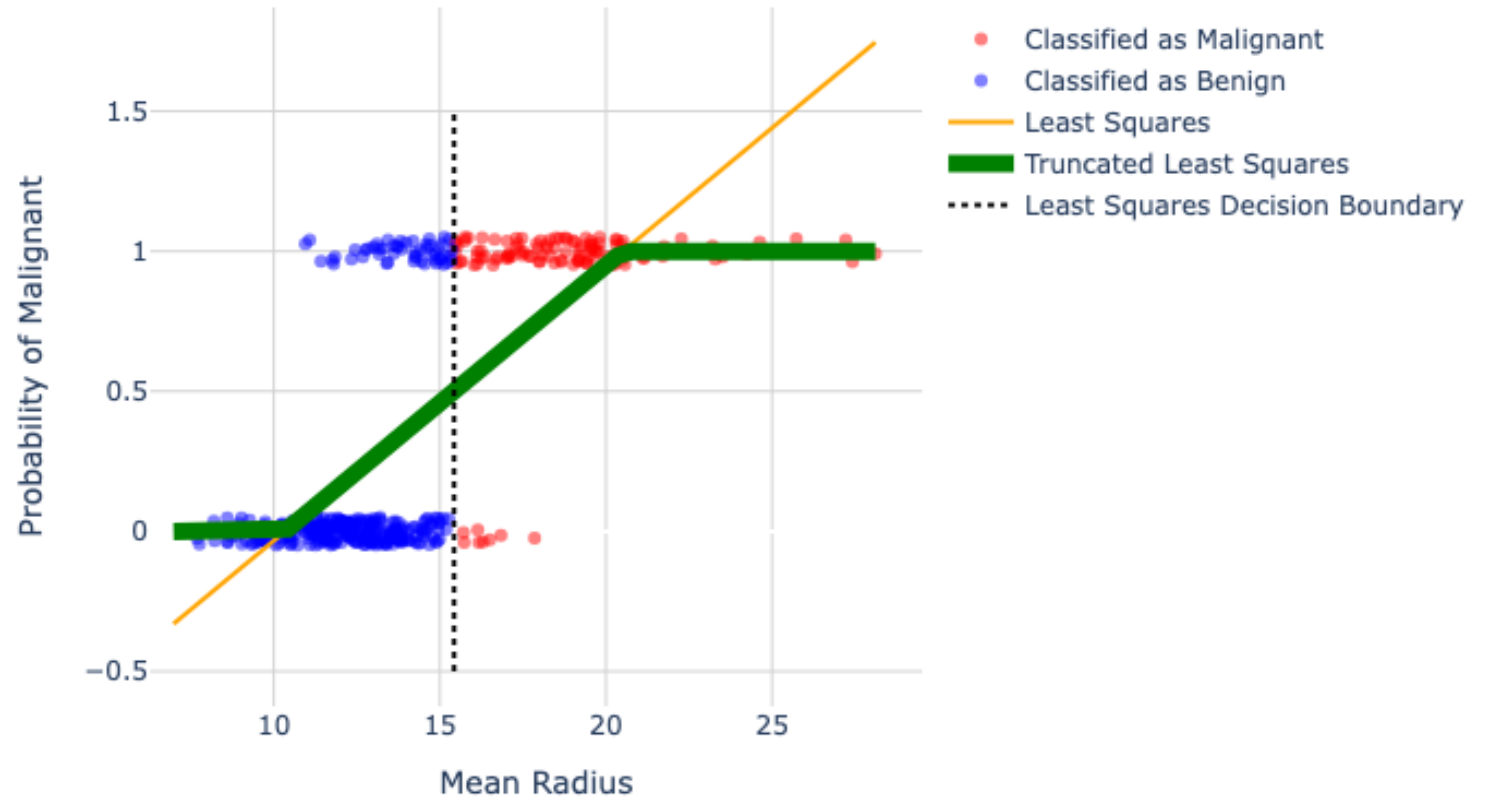


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Demo

It is difficult to interpret a model that outputs values below 0 and above 1 so we capped the output between 0 and 1.

Breast Cancer Prediction using Linear Regression





Demo

This model is also sensitive to outliers.

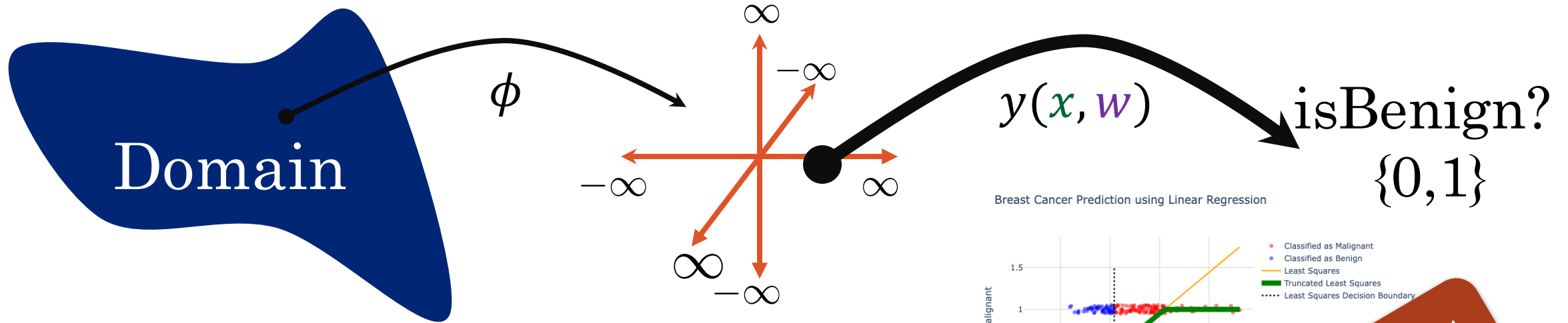
Breast Cancer Prediction with an Outlier



Classification



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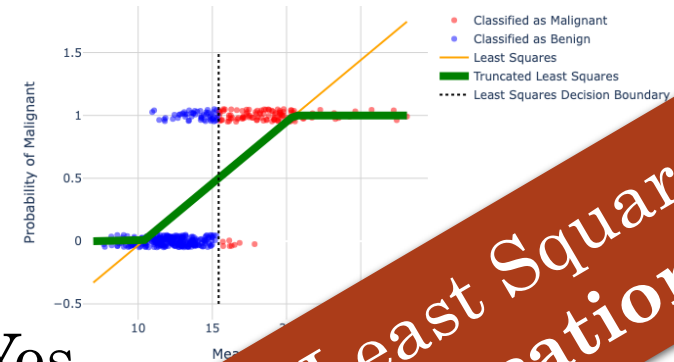
Can we just use least squares?

$$\operatorname{argmin}_w \frac{1}{2} \sum_{n=1}^N (t_n - w^T \phi(x_n))^2 + \frac{\lambda}{2} w^T w$$

Squared Error

Regularization

Breast Cancer Prediction using Linear Regression



- Yes ...
- Need a loss function (e.g., ...)
- ... to interpret model ...
- Sensitive to outliers

Don't use Least Squares for Classification

Least squares assumes
Gaussian prior, so it fails
with binary data.

Takeaway!!!

Defining a New Model for Classification

- Classification Task
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- Linear Discriminant Functions
- Discriminative Probabilistic Models
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Classification Setup

- **Goal:** Map an input vector $\mathbf{x} \in \mathbb{R}^D$ to one of the K **discrete classes** C_k where $k = 1, 2, \dots, K$.
 - $K = 2$: **Binary Classification**
 - $K > 2$: **Multi-class Classification**
- Input space is divided into decision regions. Boundaries are **decision boundaries** or **decision surfaces**.
 - $(D - 1)$ -dimensional hyperplanes within the D -dimensional input space.

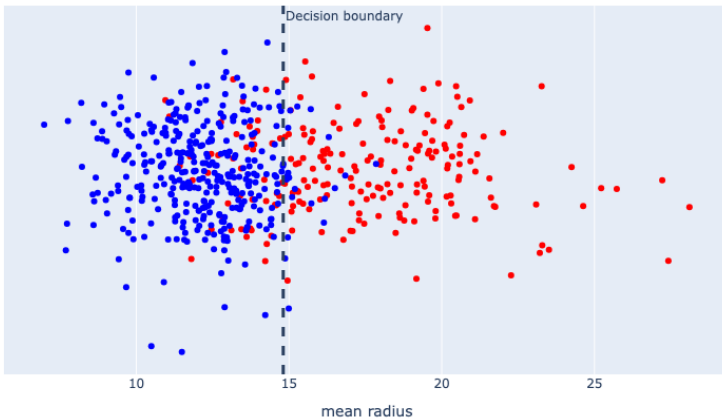
Classification Setup



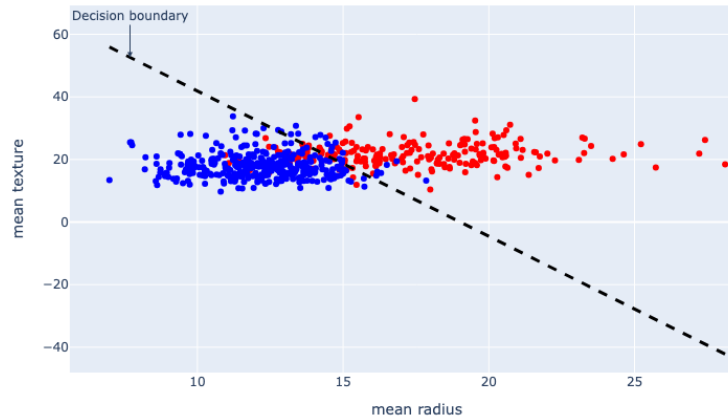
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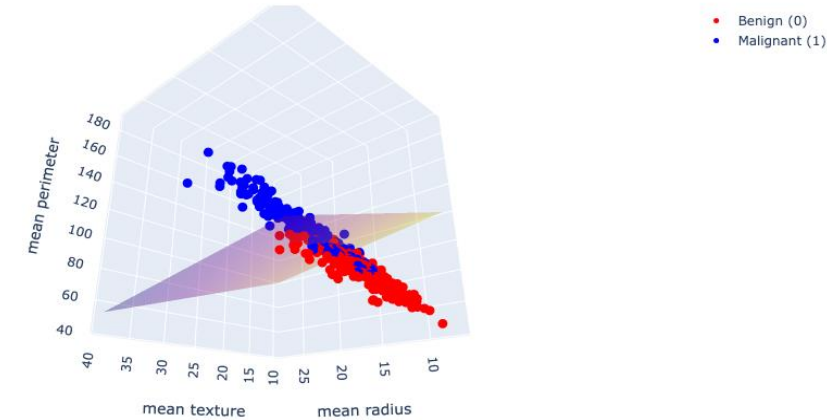
$D = 1 \rightarrow$ decision boundary is 0-D (a point)



$D = 2 \rightarrow$ decision boundary is 1-D (a line)



$D = 3 \rightarrow$ decision boundary is 2-D (a plane)



- Dataset with classes that are separable by linear decision surfaces are said to be **linearly separable**.



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3 Approaches in Solving Classification

1. Discriminant functions (direct mapping): Direct assignment to a class.

Ex: SVM (margin-based view)

2. Generative probabilistic models (model conditional probability $p(\mathbf{x}|C_k)$ and prior $p(C_k)$ then apply Bayes).

Ex: Naïve Bayes,
ChatGPT

$$p(C_k|\mathbf{x}) = \frac{p(\mathbf{x}|C_k)p(C_k)}{p(\mathbf{x})}$$

3. Discriminative probabilistic models (model conditional probability $p(C_k|\mathbf{x})$ directly).

Ex: Logistic Regression

Linear Discriminant Functions

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Discriminant Classification with Two Classes



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- **Basic representation:**

A linear function of the input vector so that $y(\mathbf{x}) = \mathbf{w}^T \mathbf{x} + w_0$.

- **Decision rule:**

$$\text{Class Assignment} = \begin{cases} C_1 & \text{if } y(\mathbf{x}) \geq 0.5 \\ C_2 & \text{otherwise} \end{cases}$$

- **Decision boundary:**

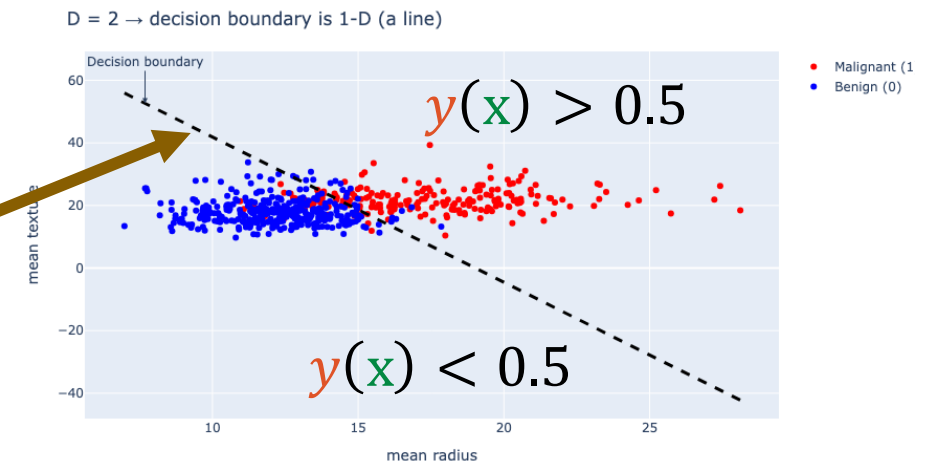
$$y(\mathbf{x}) = 0.5$$

Which is a $(D - 1)$ -dimensional surface.

$$y(\mathbf{x}) = 0.5$$

$$\mathbf{w}^T \mathbf{x} + w_0 = 0.5$$

$$x_{\text{Mean_radius}} \times w_1 + x_{\text{Mean_texture}} \times w_2 + w_0 = 0.5$$





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Geometry of Linear Discriminant Functions

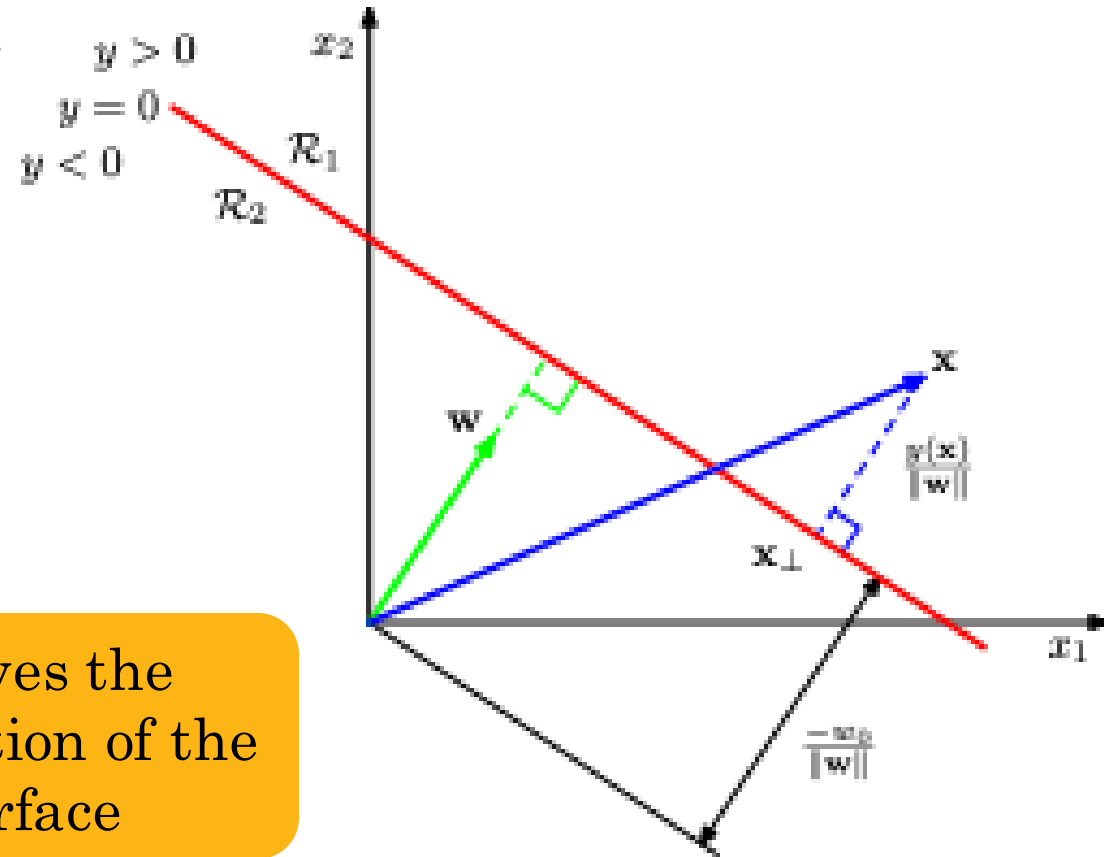
- For any two points on the line $\mathbf{x}_A$ and $\mathbf{x}_B$:

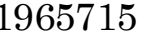
$$y(\mathbf{x}_A) = y(\mathbf{x}_B) = 0$$

$$\mathbf{w}^T(\mathbf{x}_A - \mathbf{x}_B) = 0$$

- $\mathbf{w}$ is orthogonal to every vector lying within the decision surface/boundary.

$\mathbf{w}$ gives the orientation of the surface



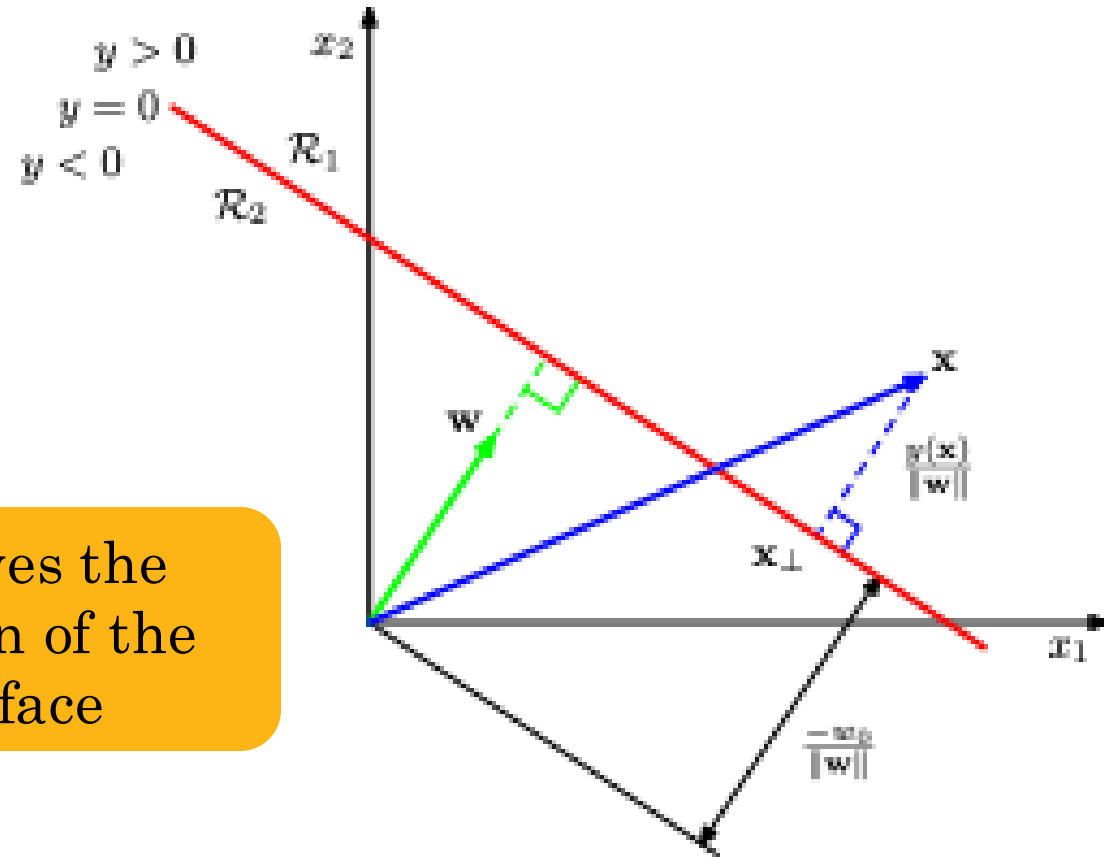


Geometry of Linear Discriminant Functions

- For every point on the decision surface, $y(\mathbf{x}) = \mathbf{w}^T \mathbf{x} + w_0 = 0$.
- The normal distance of the surface to the origin is:

$$\frac{\mathbf{w}^T \mathbf{x}}{\|\mathbf{w}\|} = - \frac{w_0}{\|\mathbf{w}\|}$$

w_0 gives the location of the surface





If our model does not have an intercept term, the decision boundary must pass through the origin.

Discriminative Probabilistic Models



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Discriminative Probabilistic Models

Logistic regression is the most widely used **discriminative probabilistic model** for **classification**.

Logistic regression models the **log-odds** of class C_1 versus C_2 as a **linear model**.

$$\text{Odds Ratio} = \frac{p(C_1|\mathbf{x})}{p(C_2|\mathbf{x})} = \frac{p(C_1|\mathbf{x})}{1-p(C_1|\mathbf{x})}$$

$$\text{Log Odds} = \ln \left(\frac{p(C_1|\mathbf{x})}{p(C_2|\mathbf{x})} \right) = \ln \left(\frac{p(C_1|\mathbf{x})}{1-p(C_1|\mathbf{x})} \right) = \mathbf{w}^\top \mathbf{x}$$

We can solve for $p(C_1|\mathbf{x})$ to obtain the form of our model.



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Deriving the Logistic Sigmoid Function

We can solve for $p(C_1|\mathbf{x})$ to obtain the form of our model.

$$\text{Log Odds} = \ln \left(\frac{p(C_1|\mathbf{x})}{p(C_2|\mathbf{x})} \right) = \ln \left(\frac{p(C_1|\mathbf{x})}{1-p(C_1|\mathbf{x})} \right) = \mathbf{w}^\top \mathbf{x}$$

$$\frac{p(C_1|\mathbf{x})}{1-p(C_1|\mathbf{x})} = \exp(\mathbf{w}^\top \mathbf{x})$$

$$p(C_1|\mathbf{x}) = \exp(\mathbf{w}^\top \mathbf{x}) (1 - p(C_1|\mathbf{x}))$$

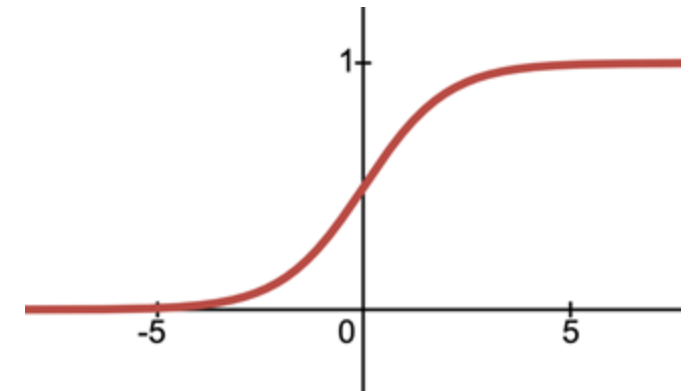
$$p(C_1|\mathbf{x}) = \exp(\mathbf{w}^\top \mathbf{x}) - \exp(\mathbf{w}^\top \mathbf{x}) p(C_1|\mathbf{x})$$

$$p(C_1|\mathbf{x}) + \exp(\mathbf{w}^\top \mathbf{x}) p(C_1|\mathbf{x}) = \exp(\mathbf{w}^\top \mathbf{x})$$

$$p(C_1|\mathbf{x}) = \frac{\exp(\mathbf{w}^\top \mathbf{x})}{1 + \exp(\mathbf{w}^\top \mathbf{x})} = \frac{1}{1 + \exp(-\mathbf{w}^\top \mathbf{x})}$$

Sigmoid Function

$$\sigma(a) = \frac{1}{1 + e^{-a}}$$



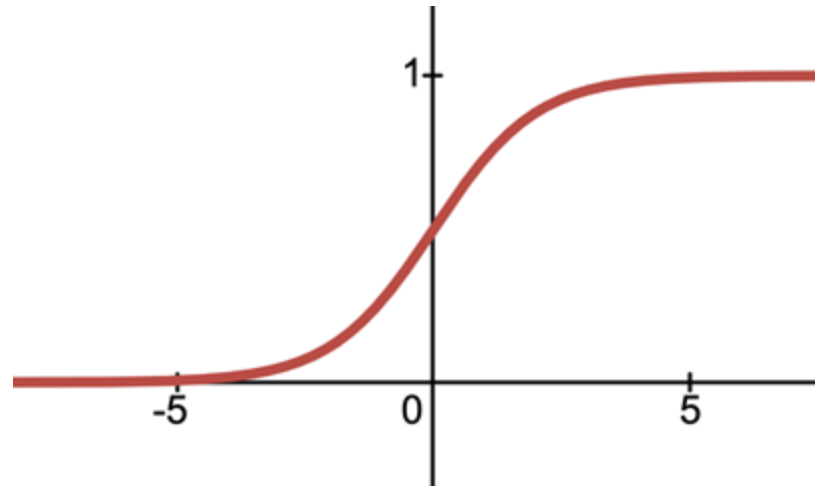
Algebra

The Sigmoid (σ) Function



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Also known as the **logistic** function.



$$\sigma(a) = \frac{1}{1 + e^{-a}}$$

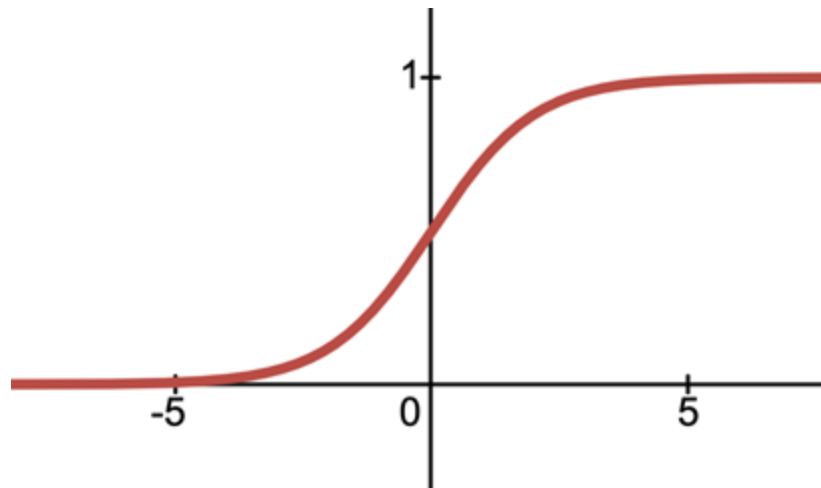
⏟
sigmoid function



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The Sigmoid (σ) Function

Also known as the **logistic** function.



The term Sigmoid means S-shaped.

A ‘squashing function’ since it maps the real axis into a finite interval.

$$\sigma(a) = \frac{1}{1 + e^{-a}}$$

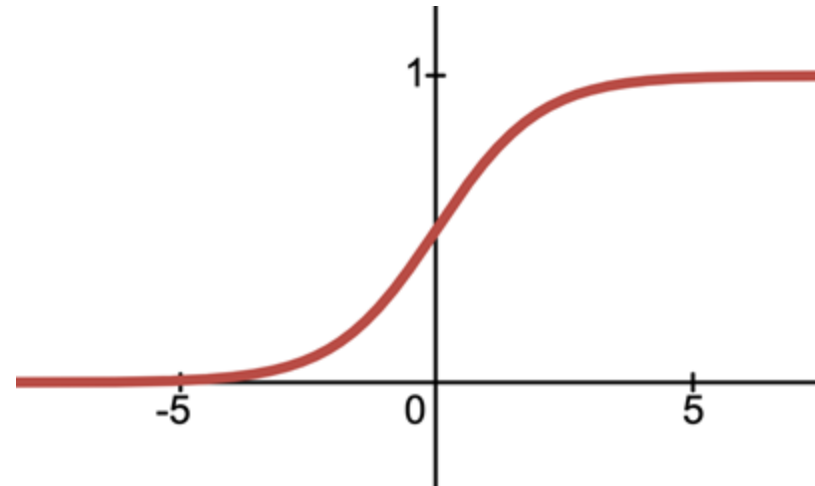
sigmoid function

Transforming the Sigmoid (σ) Function



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We can **shift** and **scale** the sigmoid function just like any other function:



$$\sigma(Aa + B) = \frac{1}{1 + e^{-(Aa+B)}}$$

([Desmos](#))

Bigger A $\rightarrow$ Horizontal shrink. Steeper!
Bigger B $\rightarrow$ Negative horizontal shift

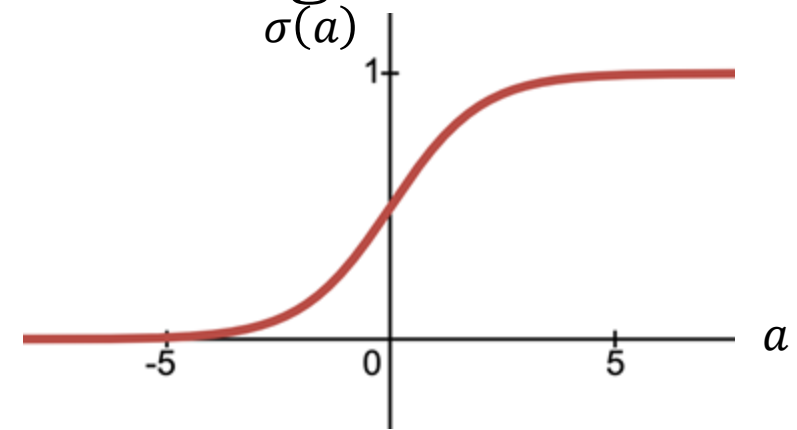
The Sigmoid (σ) Function



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The S-shaped curve is formally known as the **sigmoid function**.

$$\sigma(a) = \frac{1}{1 + e^{-a}}$$



Reflection/
Symmetry

$$1 - \sigma(a) = \frac{e^{-a}}{1 + e^{-a}} = \sigma(-a)$$

Domain
 $-\infty < a < +\infty$

Range
 $0 < \sigma(a) < 1$

Inverse

$$a = \sigma^{-1}(p) = \log\left(\frac{p}{1-p}\right)$$

Logit Function

Derivative

$$\frac{d}{da} \sigma(a) = \sigma(a)(1 - \sigma(a)) = \sigma(a)\sigma(-a)$$



Which of the following are properties of the sigmoid function, $\sigma(x)$?

Sigmoid for Classification

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Logistic Regression

- Widely used models for **binary classification**:

x = “Get a FREE sample ...”

$$\phi(\mathbf{x}) = [2.0, 0, \dots, 1.0, 0.5] \quad \Rightarrow \quad y = 1$$

1 = "Spam"
0 = "Ham"

- Models the probability $p(C_1|\phi(\mathbf{x}))$.

Why is ham good
and spam bad? ...

(<https://www.youtube.com/watch?v=anwy2MPT5RE>)

$$P(C_1|\phi(\mathbf{x})) = \sigma(\mathbf{w}^T \phi(\mathbf{x})) = \frac{1}{1 + \exp(-\mathbf{w}^T \phi(\mathbf{x}))}$$

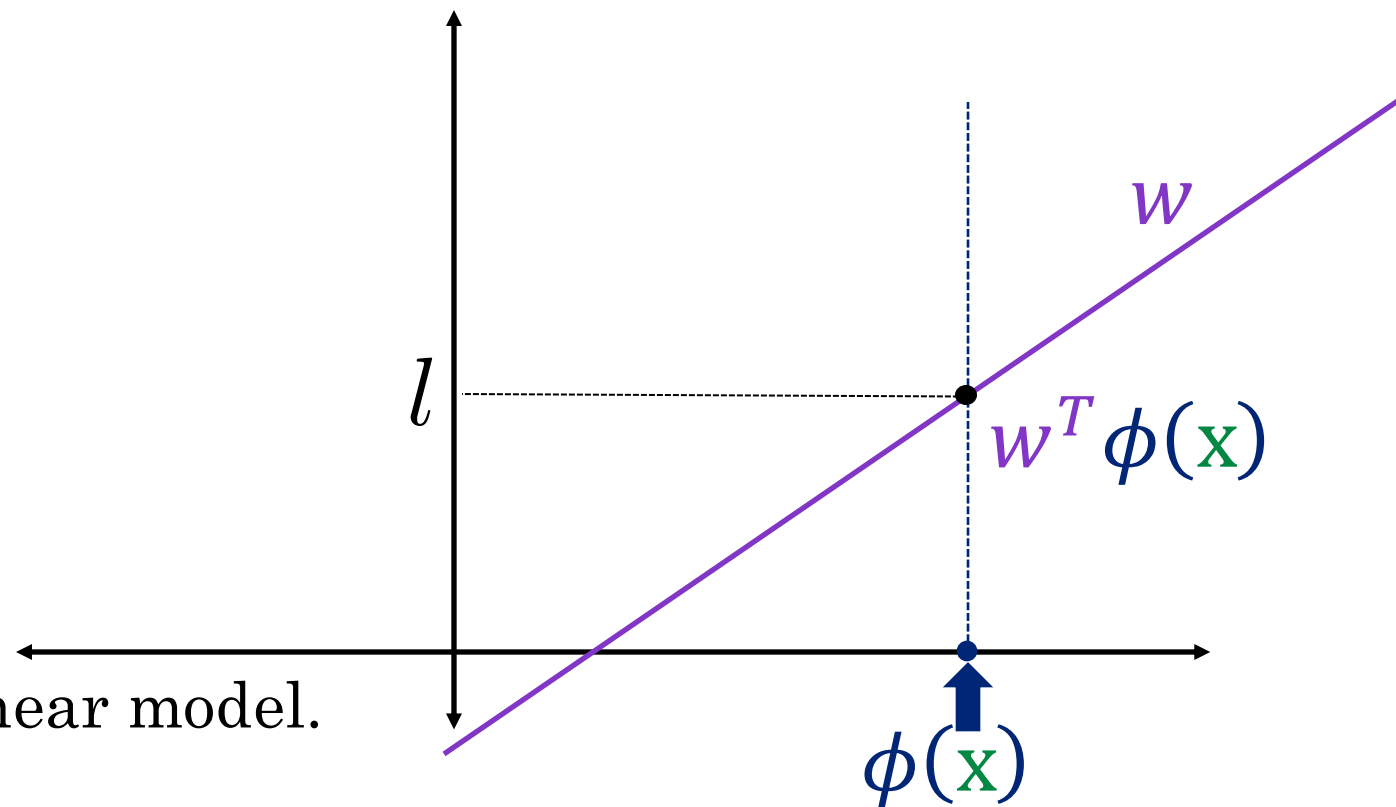
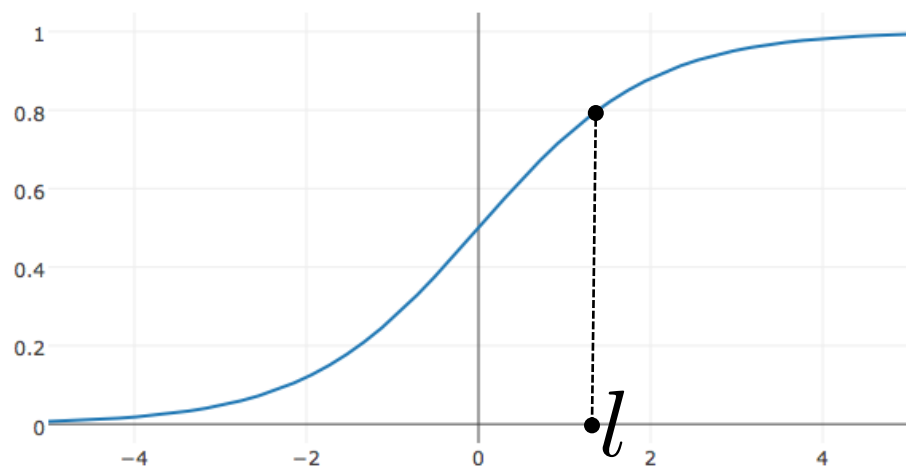
Logistic Regression



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Linear Model

$$\text{Model: } P(C_1 | \phi(\mathbf{x})) = \underbrace{\sigma}_{\text{Sigmoid}} \left(\underbrace{w^T \phi(\mathbf{x})}_{\text{Linear}} \right) = \frac{1}{1 + \exp(-w^T \phi(\mathbf{x}))}$$



Generalized Linear Model:

Non-linear transformation of a linear model.



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Logistic Regression

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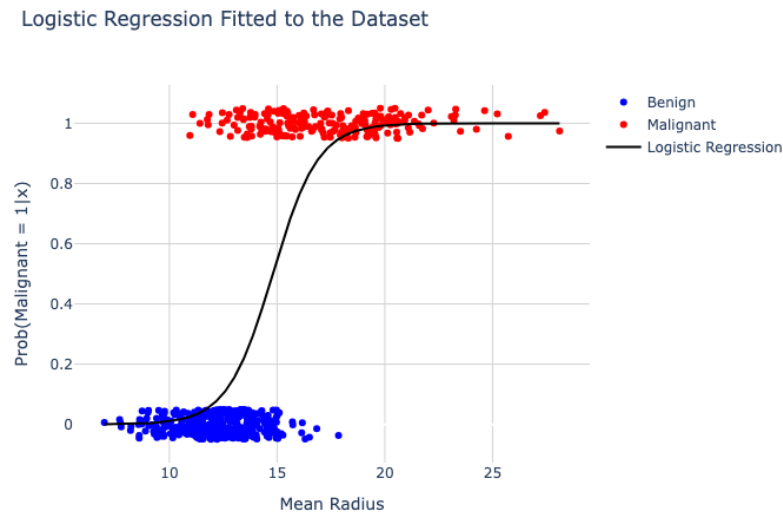
$$P(C_2|\phi(\mathbf{x})) = 1 - P(C_1|\phi(\mathbf{x}))$$

Our Cancer Classification Example



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We've discovered the (simple) **logistic regression** model! It just fits a sigmoid to the data.



$$p(C_1|\phi(\mathbf{x})) = \sigma(w_0 + w_1\mathbf{x}) = \frac{1}{1 + e^{-(w_0 + w_1\mathbf{x})}}$$

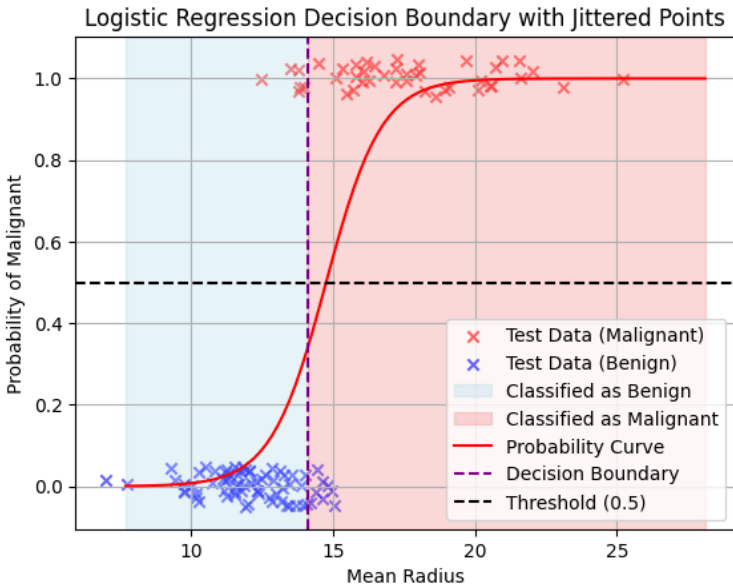
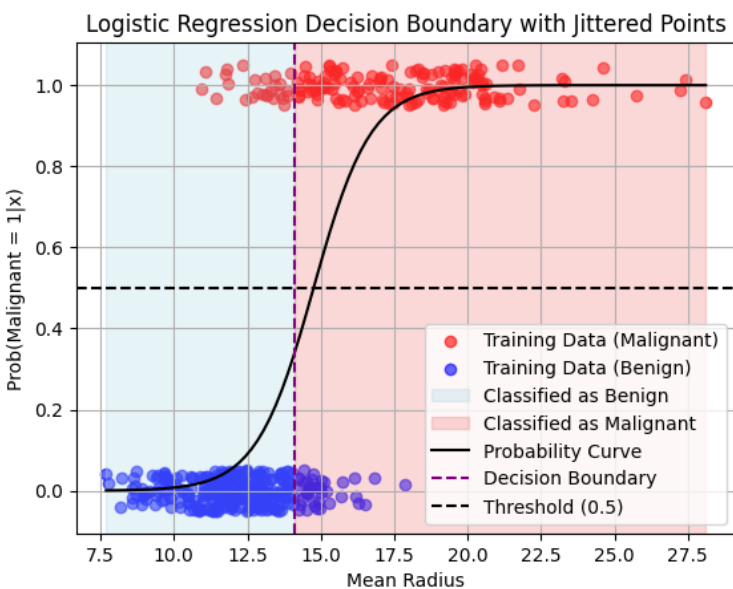
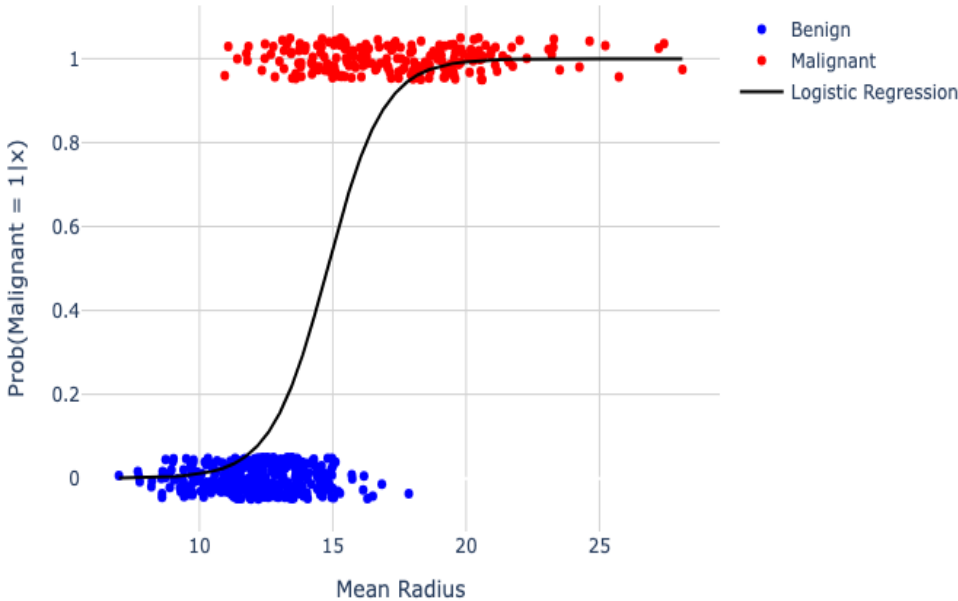
How do we identify the optimal w 's? More shortly!

Logistic Regression and Decision Boundary



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Logistic Regression Fitted to the Dataset



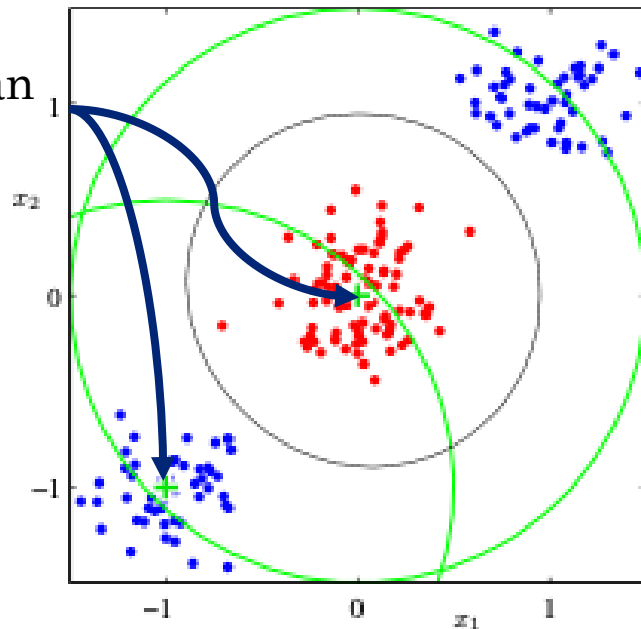
Basis Functions and Decision Boundary



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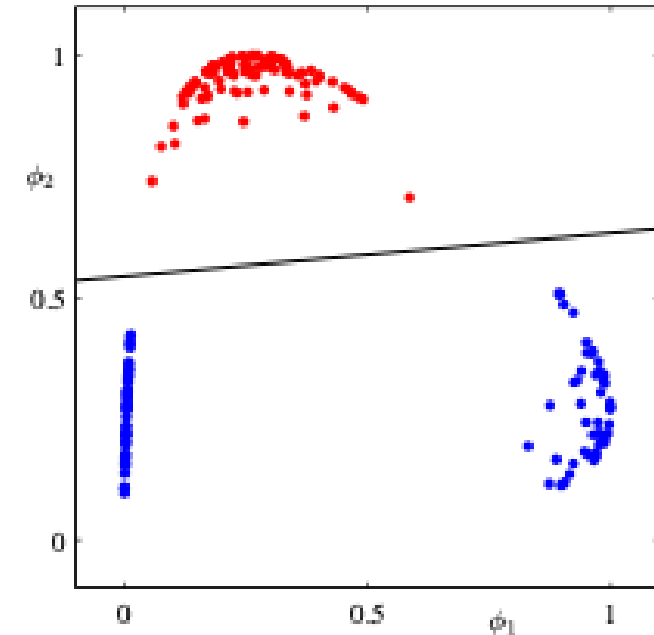
- We can still apply basis functions $\phi(\mathbf{x})$ to the classification inputs. The resulting decision boundaries will be linear in the feature space ϕ , and these correspond to nonlinear decision boundaries in the original $\mathbf{x}$.

Non-linear decision boundary in input space.



Centers of the Gaussian basis functions

Linear decision boundary in the Gaussian space.



Logistic Regression Objective

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Can We Use Squared Error
As Objective?

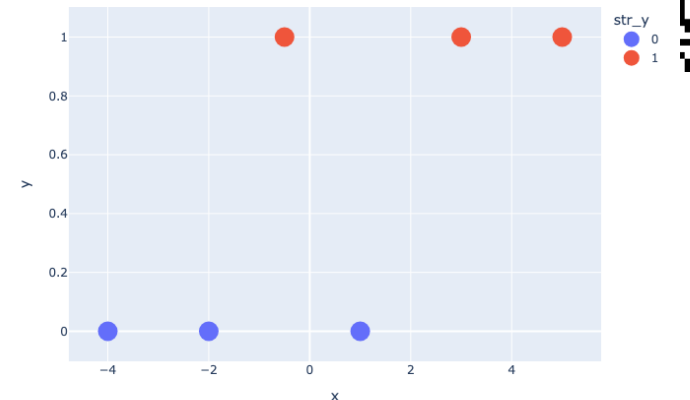
Toy Dataset: Squared Error

Logistic Regression
model:

$$p(C_1|\phi(\mathbf{x})) = \sigma(\mathbf{w}^T \phi(\mathbf{x}))$$

Assume no intercept.
So $\mathbf{x}$ and $\mathbf{w}$ are scalars.

	x	y
0	-4.0	0
1	-2.0	0
2	-0.5	1
3	1.0	0
4	3.0	1
5	5.0	1



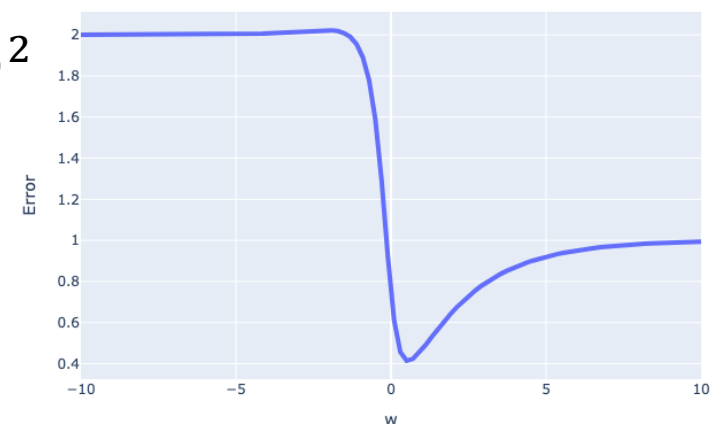
Squared Error:

$$E(\mathbf{w}) = \frac{1}{2} \sum_{n=1}^N (t_n - \sigma(\mathbf{w}^T \phi(\mathbf{x})))^2$$

The squared error
surface for logistic
regression has many
issues!



Error on Toy Classification Data



Pitfalls of Squared Error

1. Non-convex. Gets stuck in local minima.

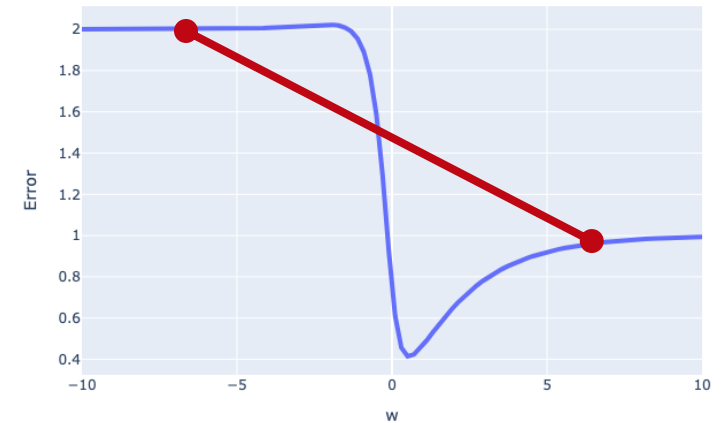
$$E(w) = \frac{1}{2} \sum_{n=1}^N (t_n - \sigma(w^T \phi(x)))^2$$



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Secant line crosses function, so $E''(w)$ is not greater than 0 for all w .

Error on Toy Classification Data



Pitfalls of Squared Error

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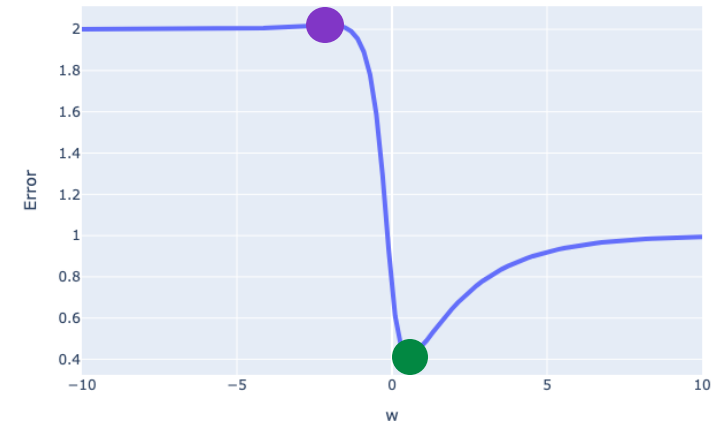
$$E(w) = \frac{1}{2} \sum_{n=1}^N (t_n - \sigma(w^T \phi(x)))^2$$



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Secant line crosses function, so $E''(w)$ is not greater than 0 for all w .

Error on Toy Classification Data



```
from scipy.optimize import minimize
```

```
minimize(ss_error_on_toy_data, x0 = 0)["x"][0]
```

0.5446601825581691

```
minimize(ss_error_on_toy_data, x0 = -5)["x"][0]
```

-10.858380927026204

Pitfalls of Squared Error

1. Non-convex. Gets stuck in local minima.

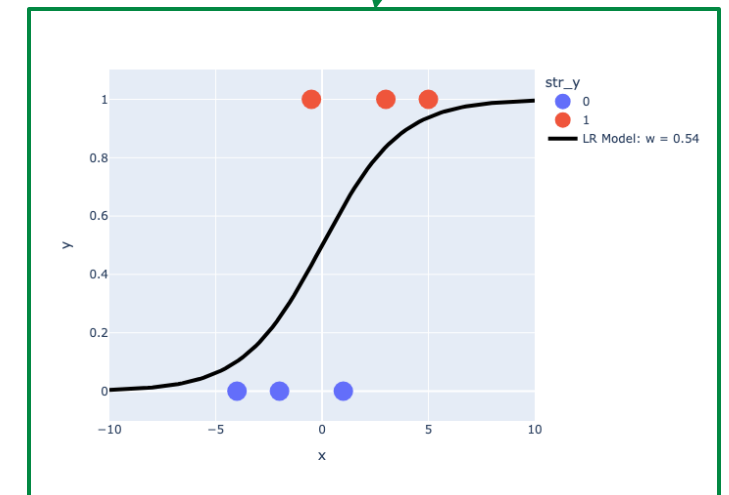
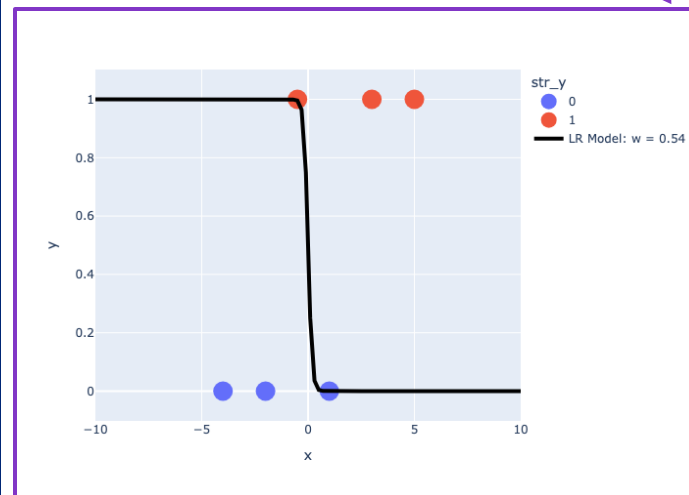
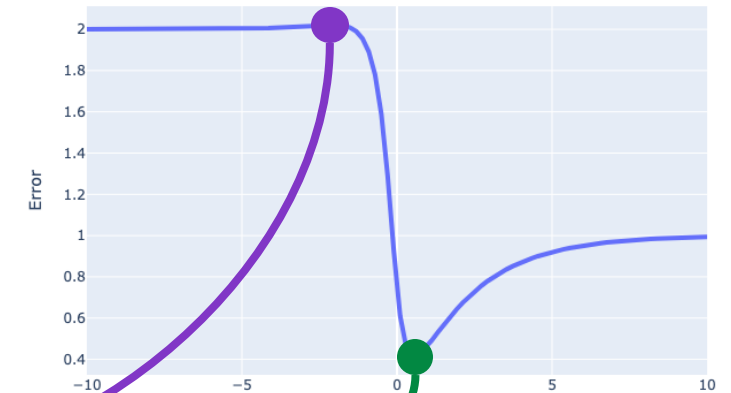
$$E(w) = \frac{1}{2} \sum_{n=1}^N (t_n - \sigma(w^T \phi(x)))^2$$



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Secant line crosses function, so $E''(w)$ is not greater than 0 for all w .

Error on Toy Classification Data





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Squared Error Gives Bounded Penalty

- Best case scenario:

Estimate matches outcome.

Squared error = $(0 - 0)^2 = (1 - 1)^2 = \mathbf{0}$. Perfect prediction!

- Worst case scenarios:

Estimated probability = 0, Actual outcome = 1.

$$\text{Squared error} = (0 - 1)^2 = \mathbf{1}$$

Estimated probability = 1, Actual outcome = 0.

$$\text{Squared error} = (1 - 0)^2 = \mathbf{1}$$

- The largest possible squared error is 1. Not a huge penalty for a big mistake!

Pitfalls of Squared Error

1. Non-convex. Gets stuck in local minima.

2. Bounded. Not a good measure of model error.

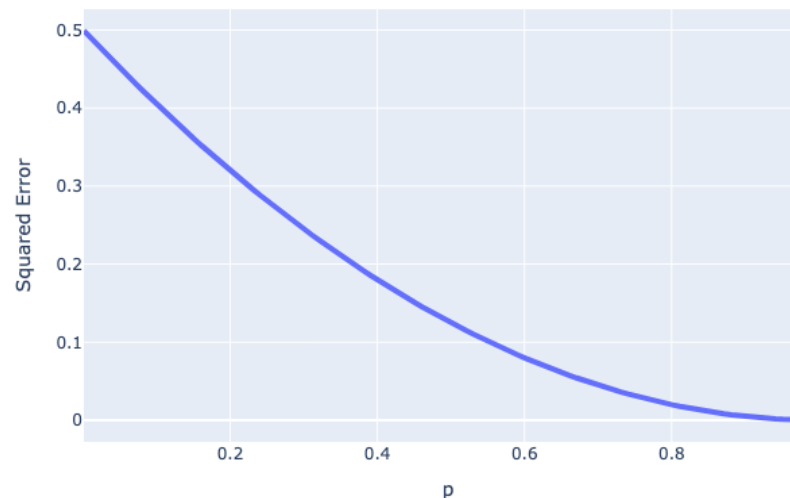
We'd like error functions to penalize "off" predictions. SE never gets very large, because both response and estimated probability are bounded by 1.

$$E(w) = \frac{1}{2} \sum_{n=1}^N (t_n - \sigma(w^T \phi(x)))^2$$



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Squared Loss for One Individual when $y=1$



If true $t = 1$ but predicted probability = 0:

$$(t - p)^2 = (1 - 0)^2 = 1$$

A New Error Function for Logistic Regression

The Logistic Regression Objective (MLE)



- In logistic regression, we can adopt **maximum likelihood estimation**.
- Define target variable $t \in \{0, 1\}$, with $t = 1$ representing class C_1 and $t = 0$ for class C_2 .
- The likelihood of t given $\mathbf{x}$ under our model $p(C_1|\mathbf{x}) = \sigma(\mathbf{w}^\top \mathbf{x})$:

$$p(t|\mathbf{w}) = \underbrace{\sigma(\mathbf{w}^\top \mathbf{x})^t}_{\text{for } t_n = 1, \text{ only this term stays}} \underbrace{[1 - \sigma(\mathbf{w}^\top \mathbf{x})]^{1-t}}_{\text{for } t_n = 0, \text{ only this term stays}}$$

for $t_n = 1$, only
this term stays

for $t_n = 0$, only
this term stays



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The Logistic Regression Objective (MLE)

- In logistic regression, we can adopt **maximum likelihood estimation**.
- Define target variable $t \in \{0, 1\}$, with $t = 1$ representing class C_1 and $t = 0$ for class C_2 .
- The likelihood of t given $\mathbf{x}$ under our model $p(C_1|\mathbf{x}) = \sigma(\mathbf{w}^\top \mathbf{x})$:

$$p(t|\mathbf{w}) = \sigma(\mathbf{w}^\top \mathbf{x})^t [1 - \sigma(\mathbf{w}^\top \mathbf{x})]^{1-t}$$

- For a dataset $\mathcal{D} = \{\mathbf{x}_n, t_n\}$, where $y_n = \sigma(\mathbf{w}^\top \mathbf{x})$ the likelihood function can be written as:

$$p(\mathcal{D}|\mathbf{w}) = \prod_{n=1}^N y_n^{t_n} (1 - y_n)^{1-t_n}$$

The Logistic Regression Objective (MLE)



$$p(\mathcal{D}|\mathbf{w}) = \prod_{n=1}^N y_n^{t_n} (1 - y_n)^{1 - t_n}$$

We can define an error function by taking the **negative logarithm** of the **likelihood**, which gives the **cross-entropy error function**.

$$E(\mathbf{w}) = -\ln p(\mathcal{D}|\mathbf{w}) = -\sum_{n=1}^N [t_n \ln y_n + (1 - t_n) \ln(1 - y_n)]$$

Cross-Entropy: Two Error Functions In One!



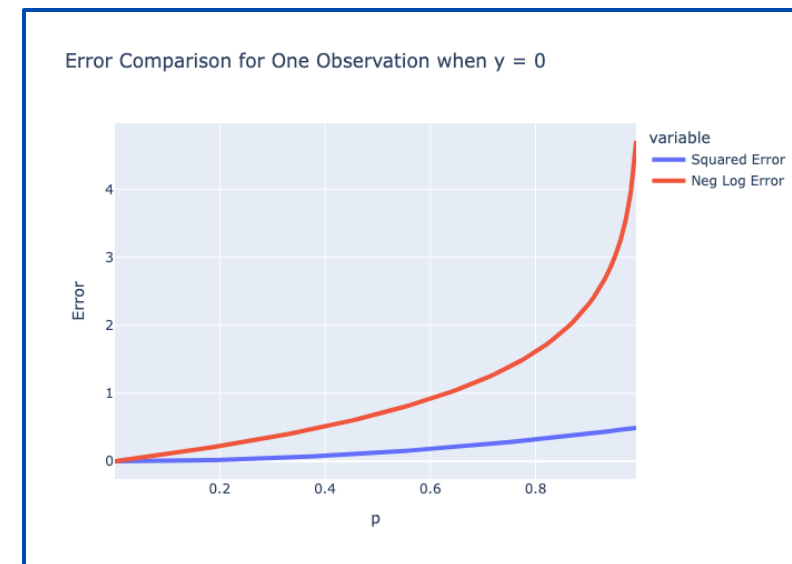
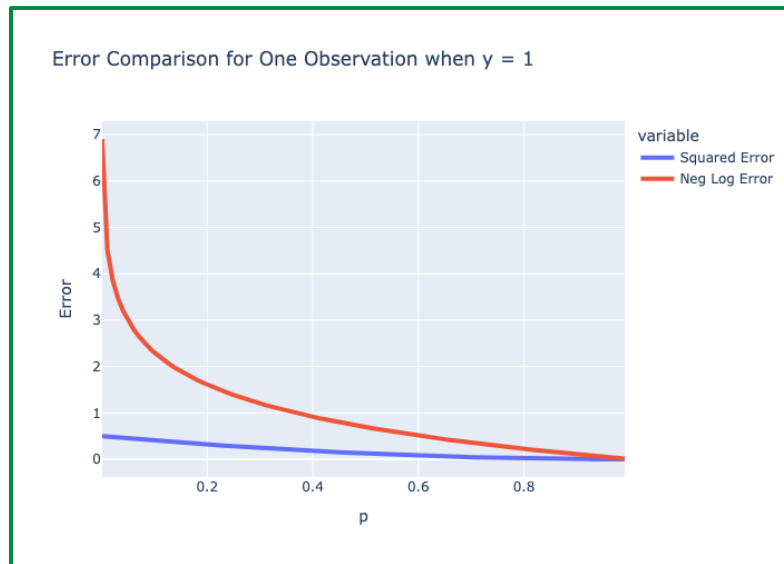
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The Cross-Entropy (CE) function is often written more compactly as the **product of the label** and **the log probability**, summed over both classes.

$$\text{CE error} = \begin{cases} -\ln y_n, & \text{if } t_n = 1 \\ -\ln 1 - y_n & \text{if } t_n = 0 \end{cases} \leftarrow -[\underbrace{t_n \ln y_n}_{\text{for } t_n = 1} + \underbrace{(1 - t_n) \ln(1 - y_n)}_{\text{for } t_n = 0}]$$

for $t_n = 1$, only this term stays

for $t_n = 0$, only this term stays



Unbounded for wrong-way certainty



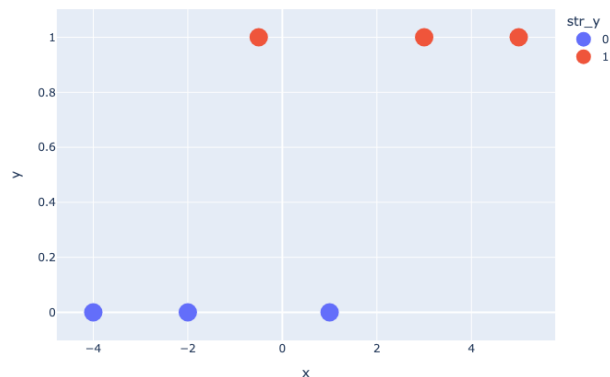
What w value will achieve 0 cross-entropy error if we train on the datapoint $x = 1, t = 1$?

Convexity By Picture



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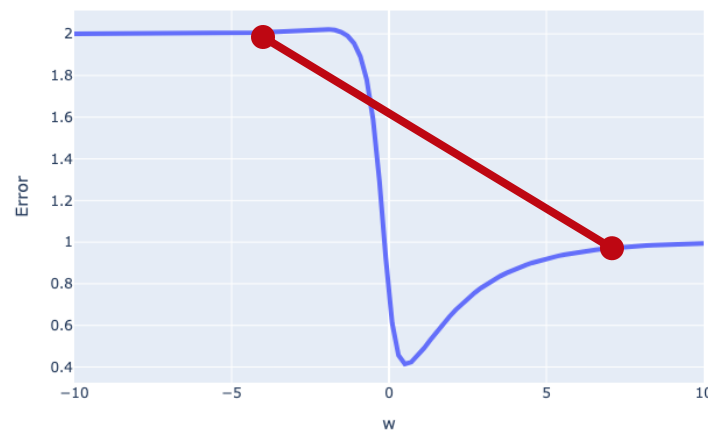
$$w = \underset{w}{\operatorname{argmin}} - \sum_{n=1}^N [t_n \ln y_n + (1 - t_n) \ln(1 - y_n)]$$



	x	y
0	-4.0	0
1	-2.0	0
2	-0.5	1
3	1.0	0
4	3.0	1
5	5.0	1

Squared Error Surface

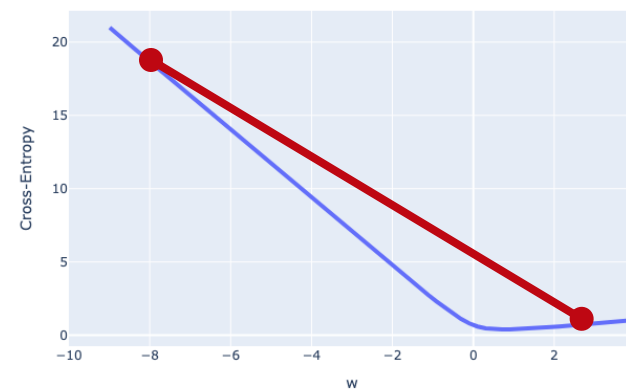
Error on Toy Classification Data



A straight line crosses the curve Non-convex

Cross-Entropy Error Surface

Cross-Entropy on Toy Classification Data



Convex!

Regularization with Logistic Regression

- Classification Task
- Defining a New Model for Classification
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- Logistic Regression Optimization
- Making Decisions



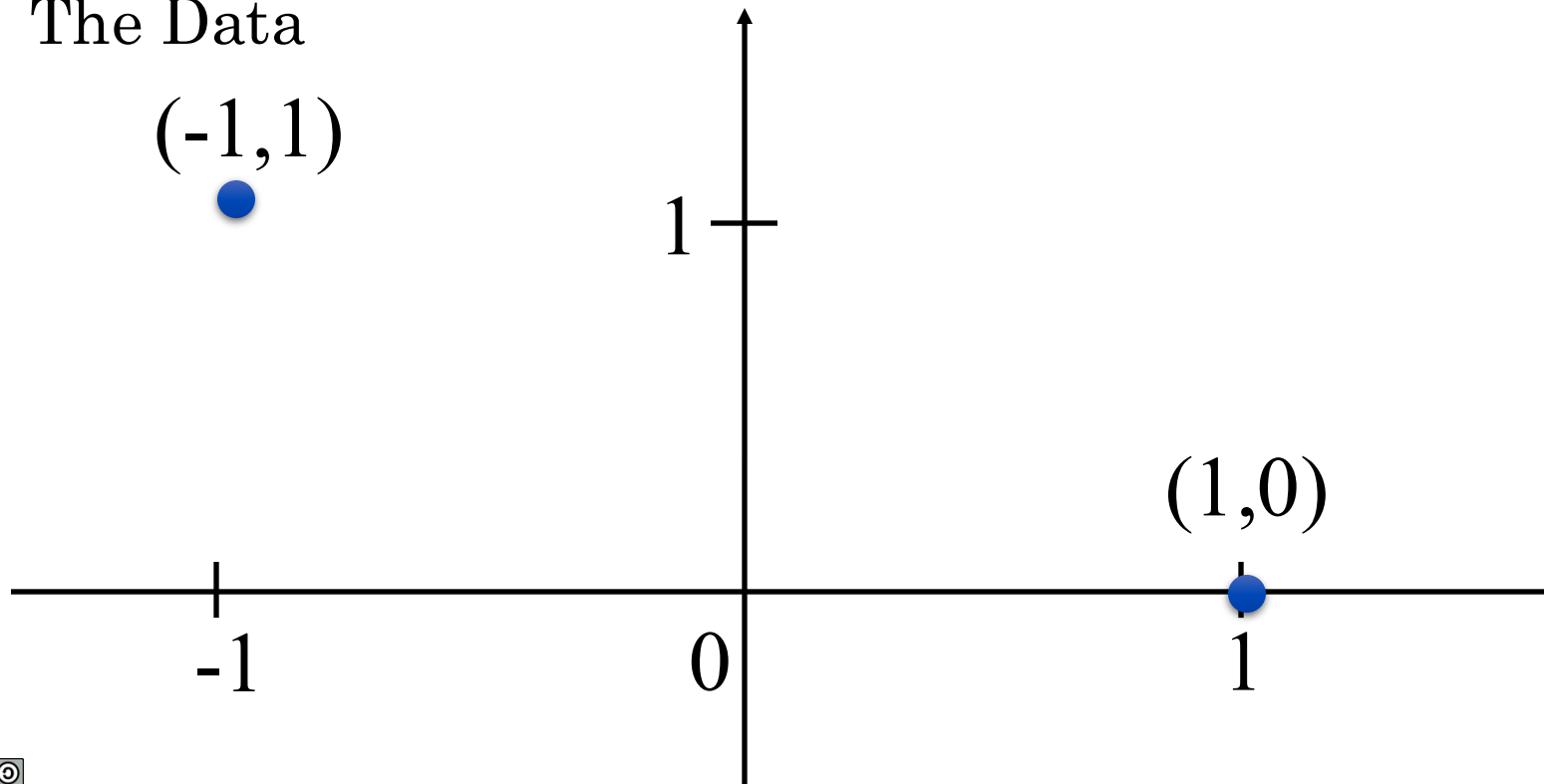
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What Is the Value of w ?

$$w = \underset{w}{\operatorname{argmin}} - \sum_{n=1}^N [t_n \ln y_n + (1 - t_n) \ln(1 - y_n)]$$

Assume $\phi(\mathbf{x}) = \mathbf{x}$

The Data

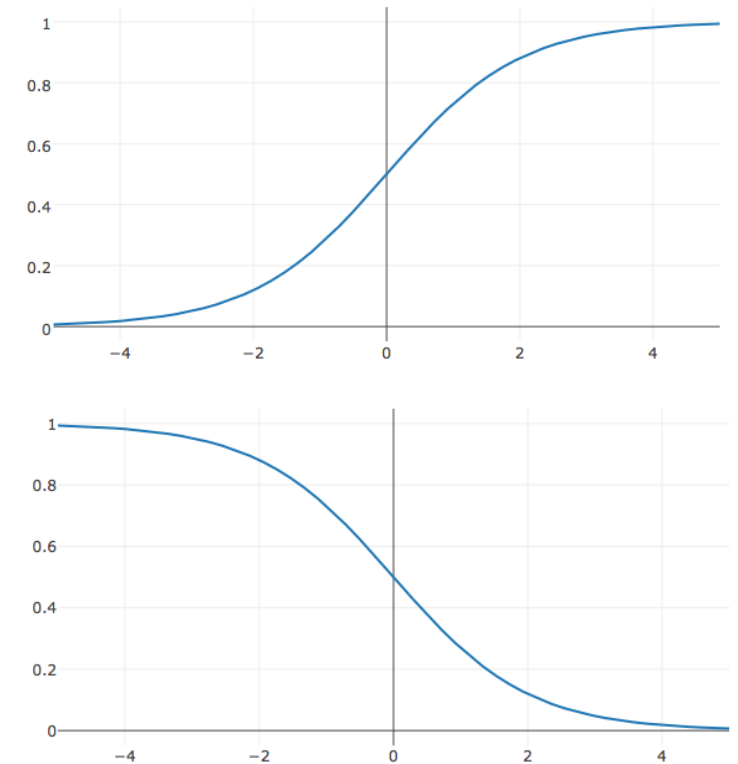


A. $w = -1$

B. $w = 1$

C. $w = -\infty$

D. $w = +\infty$



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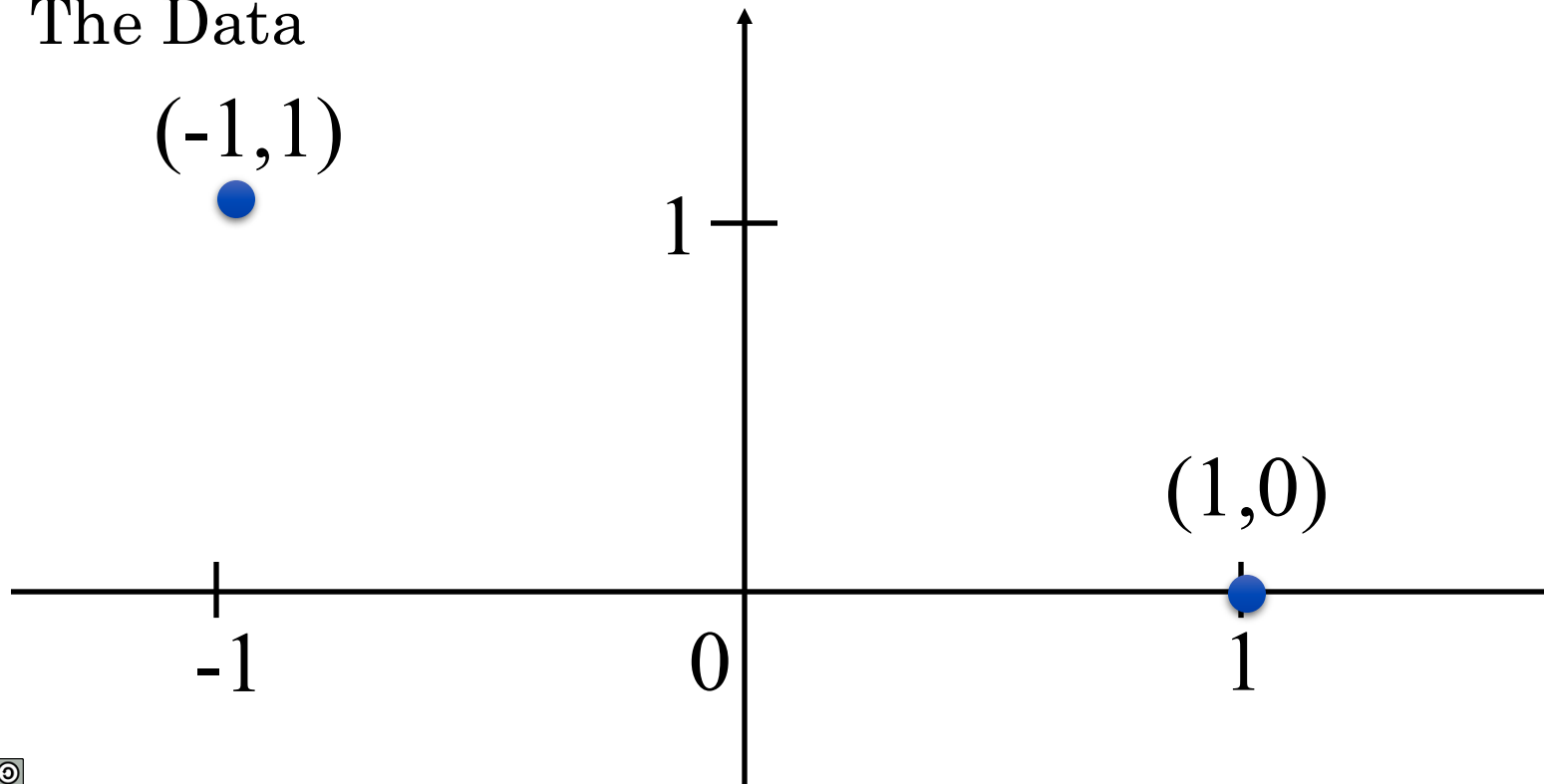
What is the best value of W ?

What Is the Value of w ?

$$w = \underset{w}{\operatorname{argmin}} - \sum_{n=1}^N [t_n \ln y_n + (1 - t_n) \ln(1 - y_n)]$$

Assume $\phi(\mathbf{x}) = \mathbf{x}$

The Data

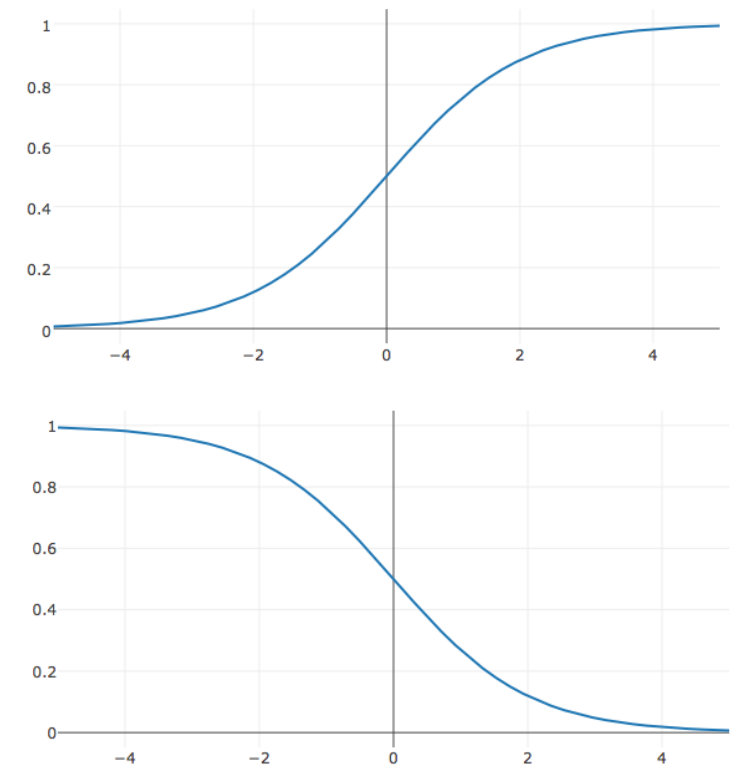


A. $w = -1$

B. $w = 1$

C. $w = -\infty$

D. $w = +\infty$



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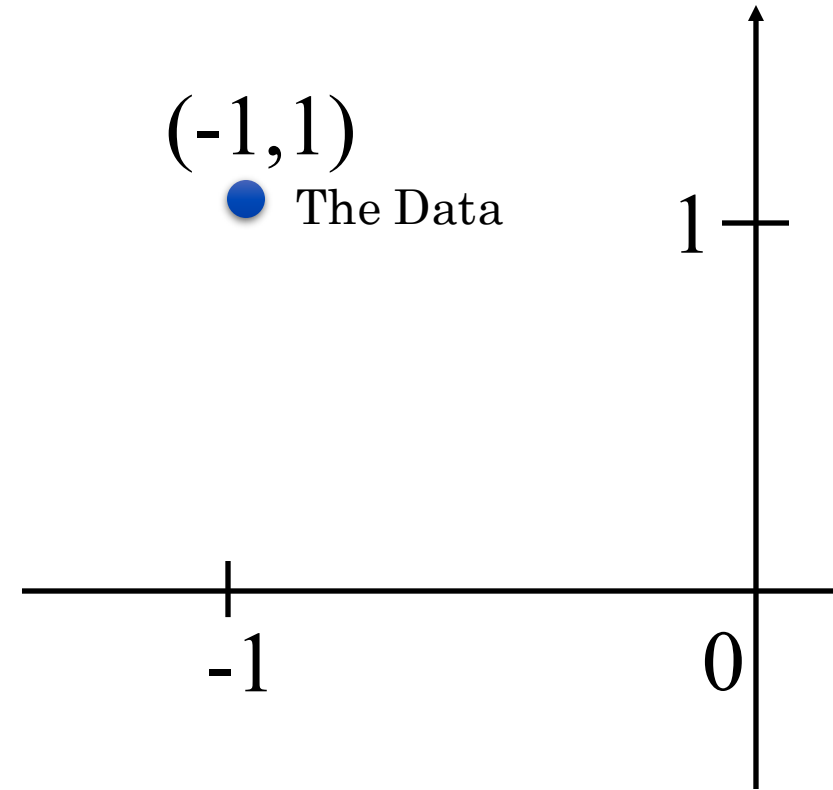
What Is the Value of w ?

$$w = \underset{w}{\operatorname{argmin}} - \sum_{n=1}^N [t_n \ln y_n + (1 - t_n) \ln(1 - y_n)]$$

For the point $(-1, 1)$:

Objective:

$$-\ln \sigma(w^T \mathbf{x}) = -\ln \sigma(-w) \Rightarrow w \rightarrow -\infty$$





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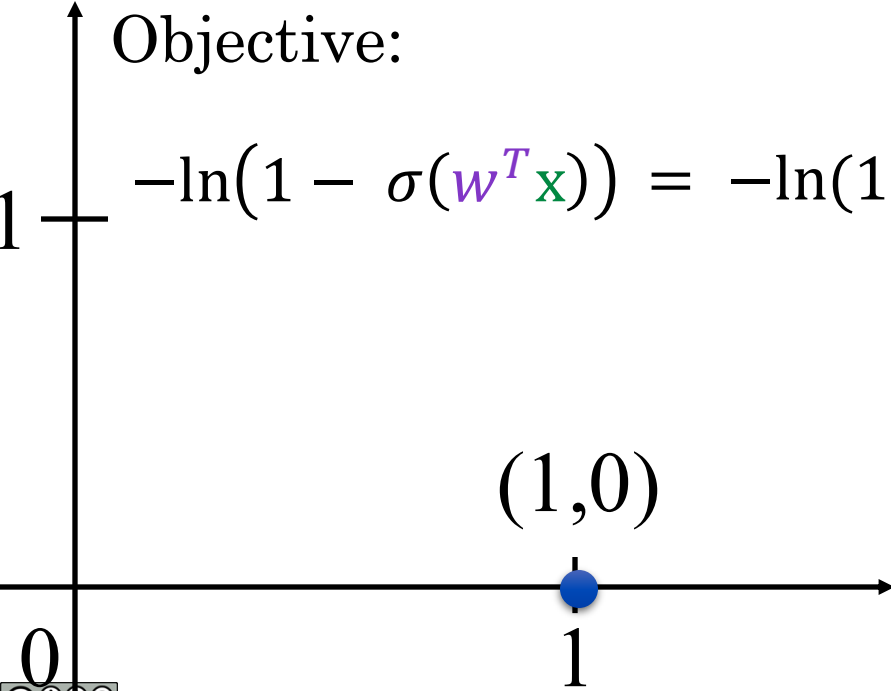
What Is the Value of w ?

$$w = \underset{w}{\operatorname{argmin}} - \sum_{n=1}^N [t_n \ln y_n + (1 - t_n) \ln(1 - y_n)]$$

For the point (1,0):

Objective:

$$-\ln(1 - \sigma(w^T \mathbf{x})) = -\ln(1 - \sigma(w)) \Rightarrow w \rightarrow -\infty$$



What Is the Value of w ?



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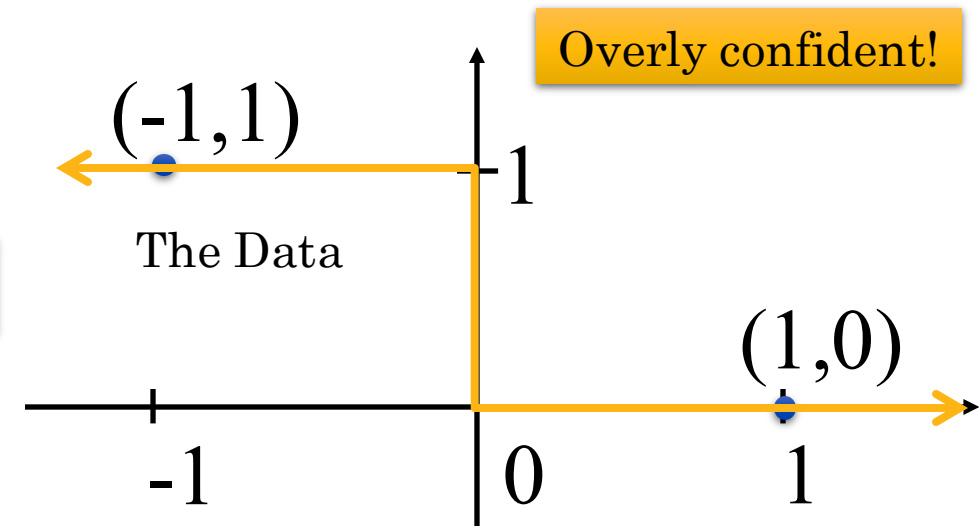
For the point $(-1,1)$:

$$-\ln \sigma(w^T \mathbf{x}) \Rightarrow w \rightarrow -\infty$$

For the point $(1,0)$:

$$-\ln(1 - \sigma(w^T \mathbf{x})) \Rightarrow w \rightarrow -\infty$$

Degenerate Solution!



Adding Regularization to Logistic Regression

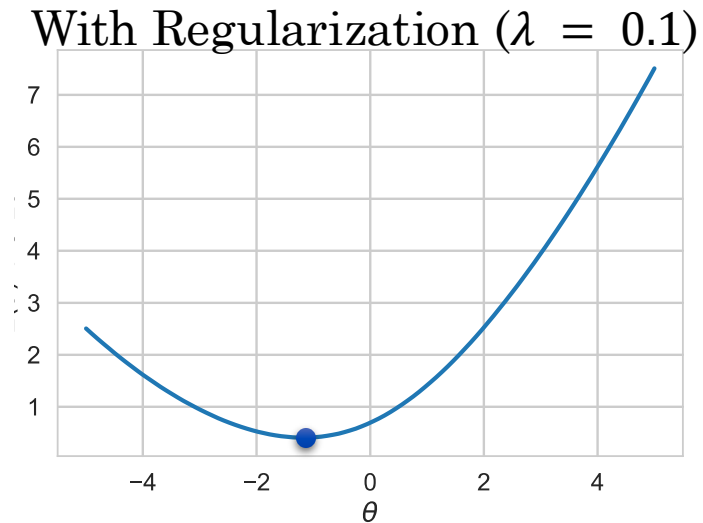
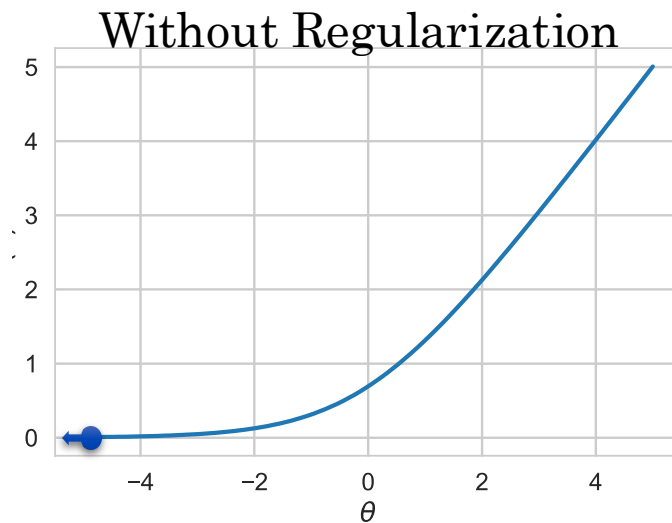


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$$\underset{\mathbf{w}}{\text{argmin}} - \sum_{n=1}^N [t_n \ln y_n + (1 - t_n) \ln(1 - y_n)] + \frac{\lambda}{2} \sum_{d=1}^D w^2$$

Prevents weights from diverging on linearly separable data.

Earlier
Example



Logistic Regression Optimization



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- Classification Task
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Logistic Regression Optimization



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$$E(\mathbf{w}) = -\ln p(\mathbf{t}|\mathbf{w}) = - \sum_{n=1}^N \left[\underbrace{t_n \ln y_n}_{\frac{t_n}{y_n} \frac{\partial y_n}{\partial \mathbf{w}}} + \underbrace{(1 - t_n) \ln(1 - y_n)}_{\frac{(1 - t_n)}{1 - y_n} \frac{\partial(1 - y_n)}{\partial \mathbf{w}}} \right]$$

Derivative wrt $\mathbf{w}$
using chain rule:

$$\frac{\partial E(\mathbf{w})}{\partial \mathbf{w}} = - \sum_{n=1}^N \left[\frac{t_n}{y_n} \frac{\partial y_n}{\partial \mathbf{w}} + \frac{(1 - t_n)}{1 - y_n} \frac{\partial(1 - y_n)}{\partial \mathbf{w}} \right]$$

Logistic Regression Optimization



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$$E(\mathbf{w}) = -\ln p(\mathbf{t}|\mathbf{w}) = - \sum_{n=1}^N \left[\underbrace{t_n \ln y_n}_{\frac{t_n}{y_n} \frac{\partial y_n}{\partial \mathbf{w}}} + \underbrace{(1 - t_n) \ln(1 - y_n)}_{\frac{(1 - t_n)}{1 - y_n} \frac{\partial(1 - y_n)}{\partial \mathbf{w}}} \right]$$

Derivative wrt $\mathbf{w}$
using chain rule:

$$\frac{\partial E(\mathbf{w})}{\partial \mathbf{w}} = - \sum_{n=1}^N \left[\frac{t_n}{y_n} \frac{\partial y_n}{\partial \mathbf{w}} + \frac{(1 - t_n)}{1 - y_n} \frac{\partial(1 - y_n)}{\partial \mathbf{w}} \right]$$

$$= - \sum_{n=1}^N \left[\frac{t_n}{y_n} \frac{\partial y_n}{\partial \mathbf{w}} - \frac{(1 - t_n)}{1 - y_n} \frac{\partial y_n}{\partial \mathbf{w}} \right]$$

$$\frac{\partial(1 - y_n)}{\partial \mathbf{w}} = - \frac{\partial y_n}{\partial \mathbf{w}}$$



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$$E(\mathbf{w}) = -\ln p(\mathbf{t}|\mathbf{w}) = -\sum_{n=1}^N \left[\underbrace{t_n \ln y_n}_{\frac{t_n}{y_n} \frac{\partial y_n}{\partial \mathbf{w}}} + \underbrace{(1 - t_n) \ln(1 - y_n)}_{\frac{(1 - t_n)}{1 - y_n} \frac{\partial(1 - y_n)}{\partial \mathbf{w}}} \right]$$

Derivative wrt $\mathbf{w}$
using chain rule:

$$\frac{\partial E(\mathbf{w})}{\partial \mathbf{w}} = -\sum_{n=1}^N \left[\frac{t_n}{y_n} \frac{\partial y_n}{\partial \mathbf{w}} + \frac{(1 - t_n)}{1 - y_n} \frac{\partial(1 - y_n)}{\partial \mathbf{w}} \right]$$

$$= -\sum_{n=1}^N \left[\frac{t_n}{y_n} \frac{\partial y_n}{\partial \mathbf{w}} - \frac{(1 - t_n)}{1 - y_n} \frac{\partial y_n}{\partial \mathbf{w}} \right]$$

$$= -\sum_{n=1}^N \left[\frac{t_n}{y_n} - \frac{(1 - t_n)}{1 - y_n} \right] \frac{\partial y_n}{\partial \mathbf{w}}$$

$$\frac{\partial(1 - y_n)}{\partial \mathbf{w}} = -\frac{\partial y_n}{\partial \mathbf{w}}$$



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$$E(\mathbf{w}) = -\ln p(\mathbf{t}|\mathbf{w}) = -\sum_{n=1}^N [\mathbf{t}_n \ln y_n + (1 - \mathbf{t}_n) \ln(1 - y_n)]$$

$$\frac{\partial E(\mathbf{w})}{\partial \mathbf{w}} = -\sum_{n=1}^N \left[\frac{\mathbf{t}_n}{y_n} \frac{\partial y_n}{\partial \mathbf{w}} + \frac{(1 - \mathbf{t}_n)}{1 - y_n} \frac{\partial (1 - y_n)}{\partial \mathbf{w}} \right]$$

$$\frac{\partial (1 - y_n)}{\partial \mathbf{w}} = -\frac{\partial y_n}{\partial \mathbf{w}}$$

$$= -\sum_{n=1}^N \left[\frac{\mathbf{t}_n}{y_n} \frac{\partial y_n}{\partial \mathbf{w}} - \frac{(1 - \mathbf{t}_n)}{1 - y_n} \frac{\partial y_n}{\partial \mathbf{w}} \right]$$

$$= -\sum_{n=1}^N \left[\frac{\mathbf{t}_n}{y_n} - \frac{(1 - \mathbf{t}_n)}{1 - y_n} \right] \frac{\partial y_n}{\partial \mathbf{w}}$$

$y_n = \sigma(\mathbf{w}^T \phi(\mathbf{x}_n))$
Need to apply chain rule

$$\frac{\partial E(\mathbf{w})}{\partial \mathbf{w}} = -\sum_{n=1}^N \left[\frac{\mathbf{t}_n}{y_n} - \frac{(1 - \mathbf{t}_n)}{1 - y_n} \right] y_n(1 - y_n) \phi(\mathbf{x}_n)$$

$$\frac{\partial y_n}{\partial \mathbf{w}} = \frac{\partial \sigma(\mathbf{w}^T \phi(\mathbf{x}_n))}{\partial \mathbf{w}} = \sigma(\mathbf{w}^T \phi(\mathbf{x}_n)) (1 - \sigma(\mathbf{w}^T \phi(\mathbf{x}_n))) \phi(\mathbf{x}_n) = y_n(1 - y_n) \phi(\mathbf{x}_n)$$



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$$E(\mathbf{w}) = -\ln p(\mathbf{t}|\mathbf{w}) = -\sum_{n=1}^N [t_n \ln y_n + (1 - t_n) \ln(1 - y_n)]$$

$$\frac{\partial E(\mathbf{w})}{\partial \mathbf{w}} = -\sum_{n=1}^N \left[\frac{t_n}{y_n} - \frac{(1 - t_n)}{1 - y_n} \right] y_n(1 - y_n) \phi(\mathbf{x}_n) \quad \text{Separating the terms}$$

$$= -\sum_{n=1}^N [t_n(1 - y_n) \phi(\mathbf{x}_n) - (1 - t_n)y_n \phi(\mathbf{x}_n)]$$

$$= -\sum_{n=1}^N [t_n \phi(\mathbf{x}_n) - \cancel{t_n y_n \phi(\mathbf{x}_n)} - y_n \phi(\mathbf{x}_n) + \cancel{t_n y_n \phi(\mathbf{x}_n)}]$$

$$= -\sum_{n=1}^N [t_n \phi(\mathbf{x}_n) - y_n \phi(\mathbf{x}_n)] = \sum_{n=1}^N [y_n - t_n] \phi(\mathbf{x}_n)$$



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We Cannot Solve Directly

$$E(\mathbf{w}) = -\ln p(\mathbf{t}|\mathbf{w}) = -\sum_{n=1}^N [t_n \ln y_n + (1 - t_n) \ln(1 - y_n)]$$

$$\frac{\partial E(\mathbf{w})}{\partial \mathbf{w}} = \sum_{n=1}^N [y_n - t_n] \phi(\mathbf{x}_n)$$

$$= \sum_{n=1}^N [\sigma(\mathbf{w}^T \phi(\mathbf{x}_n)) - t_n] \phi(\mathbf{x}_n)$$

No Closed Form
Solution

$\mathbf{w}$ is inside the σ for every
term so we cannot pull out $\mathbf{w}$.

We cannot apply logit to inverse σ :
 $\text{logit}(a + b) \neq \text{logit}(a) + \text{logit}(b)$

More on this in lectures 12 and 13 with Gradient Descent.

Making Decisions

- Classification Task
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Decisions = Posteriors + Loss

- We separate *inference* from *decision*.

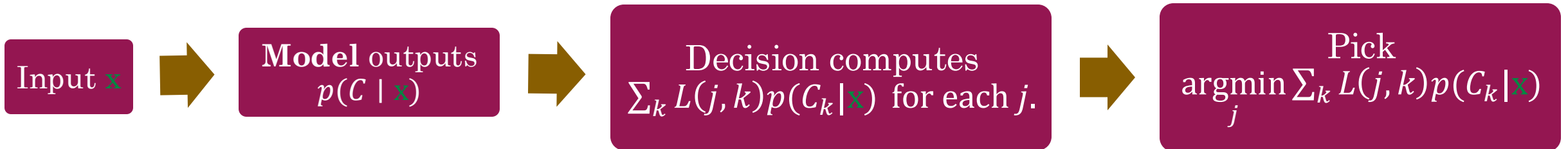
Inference gives posteriors $p(C_k|\mathbf{x})$.

The task gives a **loss matrix** L_{kj} .

L_{kj} : Loss if we predict j
when the true class is k .

The right action is the one with **minimum expected loss at this $\mathbf{x}$** .

$$\begin{array}{l} \text{Expected loss at } \mathbf{x} \text{ if} \\ \text{we choose class } j \end{array} = \sum_k L_{kj} p(C_k|\mathbf{x})$$



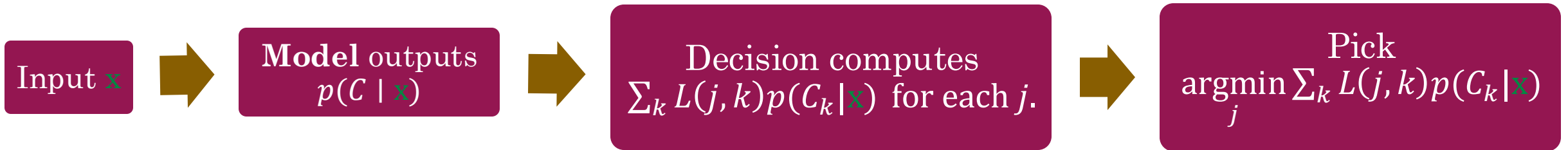


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Decisions = Posteriors + Loss

The right action is the one with **minimum expected loss** at this $\mathbf{x}$.

$$\text{Expected loss at } \mathbf{x} \text{ if we choose class } j = \sum_k L_{kj} p(C_k | \mathbf{x})$$



Change the **costs** $L \rightarrow$ decisions change immediately;
No retraining. That's why we prefer probabilities over hard labels.



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Expected Loss Example

In the example of cancer prediction, let's assume for a data sample we have:

$$p(C_0|\mathbf{x}) = 0.7 \text{ and } p(C_1|\mathbf{x}) = 0.3.$$

Assume also that the Loss matrix is given as:

	C_0	C_1
normal	0	1
cancer	100	0

If we predict normal
 $j = C_1$

$$\sum_{k=C_0} L(j, k) p(C_1|\mathbf{x}) = 100 \times 0.3 = 30$$

If we predict cancer
 $j = C_0$

$$\sum_{k=C_1} L(j, k) p(C_0|\mathbf{x}) = 1 \times 0.7 = 0.7$$

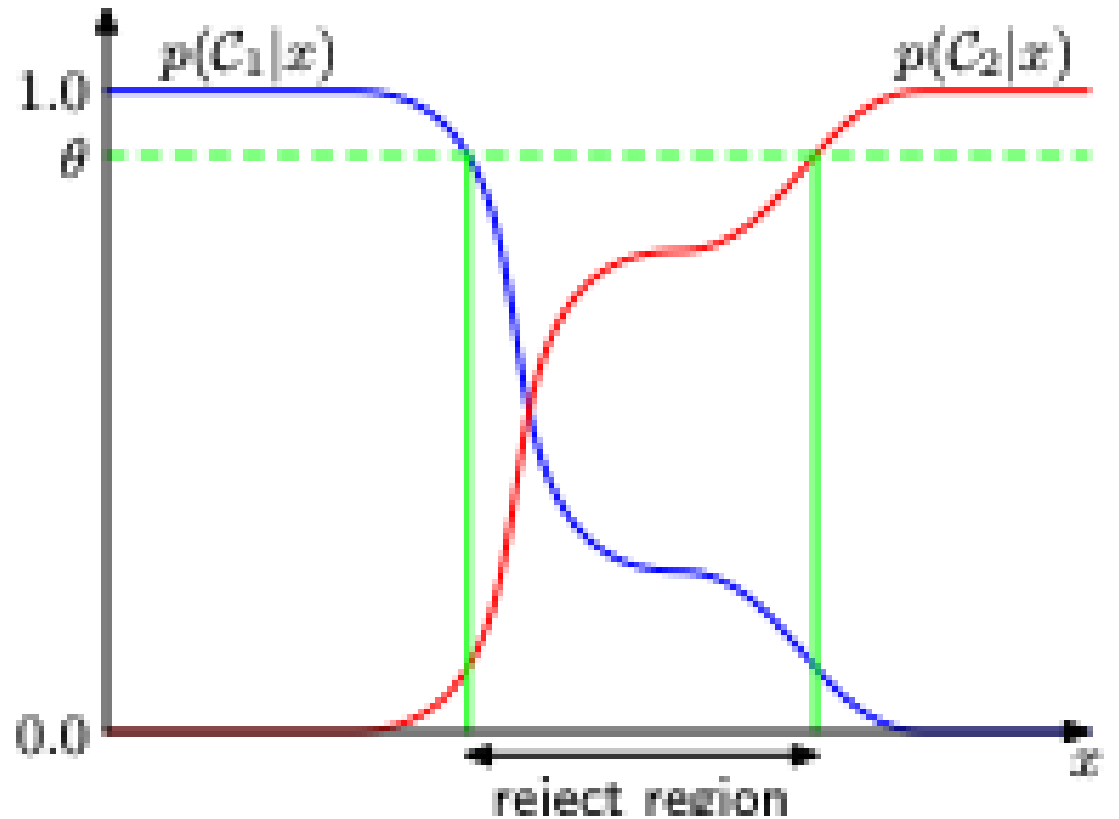
$0.7 < 30$ and predicting cancer gives smaller loss so we predict the class of cancer.

The Rejection Option



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- Reject decisions when max posterior probability $\leq$ threshold θ .
- Control error rate vs. rejection fraction.



Lecture 10

Logistic Regression (1)

Credit: Joseph E. Gonzalez and Narges Norouzi

Reference Book Chapters: Chapter 5 Sections 5.1, 5.2(up to 5.2.5), 5.4 (up to 5.4.4)